

Crystallography News

British Crystallographic Association



Issue No. 178 September 2026

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Spring and Early Summer Meetings 2026

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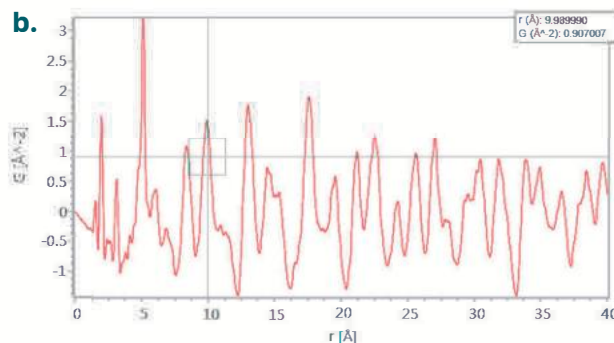
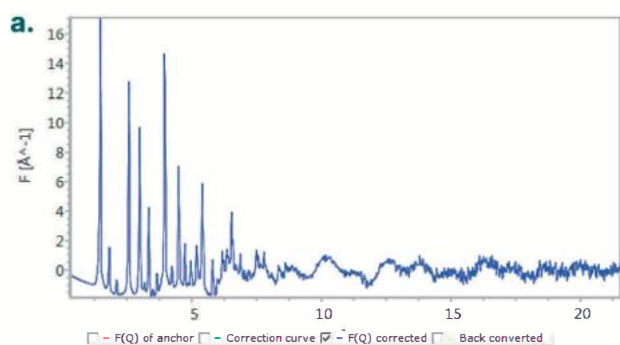
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This month's cover:

The Spring Meeting and the 2026 Bernal Lecturer, Alison Noble (Oxford), as well as the late Professor Pauline Harrison CBE (Sheffield).



From the President



MY 5-year old son has developed an interest in flint. It began practically: he found some sharp pieces, discovered they would slice through plant stems, and worked his way through a good deal of one of the University's border displays. Once I had (literally) managed to drag him away, he informed me that flint

was a crystal. I said yes, it was many very tiny crystals of quartz. He said it was just one. This went back and forth for some time with no movement on either side, until I gave up and, tempting as it was to bring up the subject of the definition of a single crystal, mosaicity, etc., changed the subject.

I have since learned that there is a proper term for the case of flint (apologies to geologically literate readers... you may skip the rest of the paragraph, or roll your eyes in despair...or both!). Apparently rocks whose crystals are too small to be made out individually by eye/light microscope are cryptocrystalline. Of course, this is key to the most functional of flint attributes – the wonderfully sharp edge on breaking. There is a related specimen on the shelf at home, an agate sample from St Egidien in the Zwickau district in Germany, a famous location for volcanic agates. It is mostly cryptocrystalline quartz as well, which is a bit easier to point out than on the flint, with the clear colour variation and banding from inclusions.

Arguments about what should count as a crystal have a longer history than this family version of course, which brings me neatly to John Finney's newly published biographical memoir of Alan Mackay, (*Biogr. Mem. Fell. R. Soc.* 20260006, ahead of print version), which I thoroughly recommend.

Elsewhere, the first round robin on structure analysis from 3D electron diffraction data caught my eye (*Gemmi et al., IUCrJ* 13, 198–210 (2026)). Three samples (epidote, natrolite, and beam-sensitive ibuprofen), with no indication of their composition, went to seven labs with different microscopes, accelerating voltages, detectors and data collection modes. All labs succeeded in determining the composition and structure of the samples, to an impressive level of precision and accuracy. One particularly nice feature of this work is that it comes in large part from a collaborative PhD student training network (EU project NanED, GA 956099).

Crystallography has a good record of round robins of this kind. As a student I was fascinated by the Commission on Powder Diffraction's Rietveld refinement round robin on PbSO_4 (*Hill, J. Appl. Cryst.* 25, 589–610, 1992). Participants were given the same X-ray and neutron data sets, and followed their own recipes and strategies. What that demonstrated was that a Rietveld refinement is not only a highly multivariate least-squares fit; it is also a test of physical and chemical reasoning. Decisions about the background, the peak-shape model, restraints and, above all, when to stop adding parameters are not settled purely by the figures

of merit. They are informed by asking whether the resulting structure, and sample-related parameters, are chemically and physically plausible. The companion study on the trickier problem of monoclinic ZrO_2 (*Hill and Cranswick, J. Appl. Cryst.* 27, 802–844 (1994)) made the same point and also gave valuable quantitative information on the quality of results from different techniques and geometries on the same material. That was a useful thing to learn early, and it remains one of the parts of the method I find most interesting. Very many happy hours have been spent digging out and perusing other data analysis related papers too since my student days and I am always very grateful that the crystallographic community is so good at detailing these issues, for better or for worse, rather than brushing them under the carpet!

Shortly before going to print we received the sad news that Kenneth Muir (Glasgow) has passed away, please look out for an obituary in a future issue. I would also like to point you those obituaries which can be found later on in this current issue. These remember Pauline Harrison (who made pivotal contributions to studies of biological iron storage), Sax Mason (a neutron specialist who undertook some of the earliest work on insulin and lysozyme), Miroslav Papiz (whose dedication to X-ray facility developments enriched a plethora of important studies) and Anthony Powell (who, aside from his many energy-related research achievements, I fondly remember as a kind and motivating community leader from my early student days onwards).

I hope by the time you read this you will have had an enjoyable (and not too hot!) late summer and I look forward to next issue's updates on the imminent IUCr Congress and General Assembly in Calgary.

Alex Gibbs,
St Andrews



Agate from St Egidien (Saxony)

BCA Council 2026

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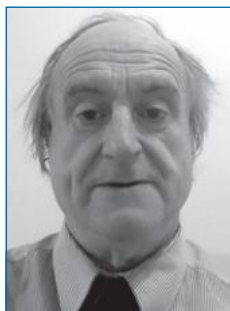


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(The dates in parentheses indicate the end of the term of office).

Full committee details on the BCA website
www.crystallography.org.uk

From the Editor



IT is a great pleasure to bring you the September 2026 issue of *Crystallography News* which features a preliminary programme for the 2027 Spring Meeting. We also have reports on the many excellent sessions at the BCA Spring Meeting in Leeds this year, including those organised by Early Stage Crystallographers Group (ESCG), the Industrial

Group (IG), the Physical Crystallography Group (PCG) and the Chemical Crystallography Group (CCG). Due to the constraints of time and space in the last issue of *Crystallography News*, we were only able to publish the reports of the BSG sessions. I am extremely grateful to everyone who sent me or uploaded accounts of sessions they attended, as well meeting photographs and in some cases incredibly detailed notes!

We then have our regular news section on the very interesting things going on at the CCDC and this is followed by a short report on the 2026 Bernal Lecture at Birkbeck which was given by **Alison Noble** (Oxford) on "Artificial intelligence in science: a perspective from medical ultrasound imaging." I am sure members will agree that this meets the remit of structural biology in its widest sense and the application of AI to any scientific discipline is one which we cannot shy away from. We then have a report on the London Structural Biology Club Summer Meeting which was organised by **Ben Stieglitz** (QMUL) at the Queen Mary Innovation Centre in early June. I am very sad to say that we then have a number of obituaries for people who were well-known to many members and the few paragraphs which I have dedicated to them hardly do them justice at all, but where possible, links are given to comprehensive obituaries published elsewhere. We then close with our regular Puzzle Corner and Meetings of Interest sections.

In the last issue I described my first steps in chemical crystallography using the GUI written by the former BCA President Richard Cooper (Oxford) for running the program CRYSTALS [1]. The program's curator, Dave Watkin (Oxford) contacted me afterwards with the following interesting comments: "*Richard had many excellent ideas, including linking CRYSTALS to CCDC's MOGUL, which of course requires that the user has a MOGUL license. MOGUL is a data base of fragment geometries in much greater detail than those listed in International Tables. The link compares the environment around each atom with a similar environment in the data base, and rates the match with a single score for each atom. The molecular geometry for atoms with a poor score is listed. If the user feels that there is an issue with the model, geometric restraints can be generated for the idealised moiety.*"

In my editorial I also discussed what, to me, was a new concept, namely the idea of using reversals in parameter refinement as an indication of convergence. Dave elaborates "*The idea came, of course, from my hero of Crystallographic Programming – John Rollett. He never received the full*

recognition he deserved, probably because he never got round to publishing his brilliant ideas. As a refinement moves towards convergence, few of the shifts change sign in successive cycles. 50% reversals are expected when the shifts are tiny and simply due to finite precision arithmetic and so doesn't always indicate an unsatisfactory convergence. More interesting is when the least squares oscillates with quite large shifts from one side of a minimum to the other. The attentive user should notice this. SHELX evades this problem by always including a Marquardt modifier to the normal equations, unless the user remembers to turn it off for the final cycles in order to get proper standard uncertainties. CRYSTALS uses the opposite strategy of only applying it to the troublesome parameters when the user sees a need. There must be squillions of other good ideas, fading like the statue of Ozymandias into the sands, which will one day be rediscovered."

"Our village is full of young folk involved with AI, but all in banking or finance. I have little fear about AI in science, except that it may have fallen into the hands of folk keen to get a trendy publication, but without any real discriminatory powers. As I understand it AI organises what we know. As scientists we need to recall [the former US Secretary of Defense] Donald Rumsfeld's sometimes ridiculed comment:"

"... as we know, there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are also unknown unknowns – the ones we don't know we don't know."

"Nail on the head, and all AI (at the moment) can do is tell us about the things we think we know."

I agree with Dave and might add that one of the challenges for scientists will be extracting physical understanding from the huge amounts of data which AI can handle better than the human mind and, for example, will biologists ever have the psychic power to gaze at an amino acid sequence and draw the correct 3D structure!?

I must thank Dave for replying with so much consideration and thought-provoking detail. Do read on for lots more interesting news items in the rest of this issue.

Jon Cooper,
UCL (Emeritus)

References

[1] Betteridge, P. W., Carruthers, J. R., Cooper, R. I., Prout, K. and Watkin, D. J. (2003). CRYSTALS version 12: software for guided crystal structure analysis. *J. Appl. Crystallogr.* **36**, 1487.



An AI adaptation of the logo for the once well-known XTAL program suite (<https://xtal.sourceforge.net/ozy.html>) which was based on the poem Ozymandias by Percy Bysshe Shelley. Given the finite number of free trials allowed at the Magic Hour website and the great antiquity of the editor's laptop, maybe this picture should be described as a one-hit wonder!

Spring Meeting 2027

University of York

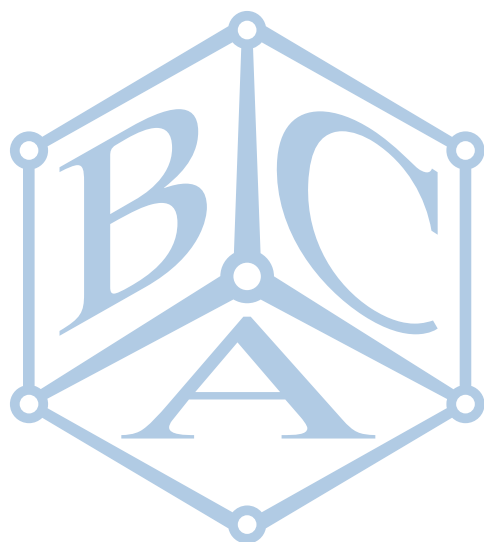
Monday 22nd March 2027

EARLY STAGE CRYSTALLOGRAPHERS GROUP (ESCG)

- 11:00-18:00 **PX001, Exhibition Centre**
The ESCG satellite meeting is an opportunity for all early-stage crystallography researchers, from across the BSG, CCG, PCG and IG, to present their work in a supportive and friendly environment, which will be run by fellow early career scientists.
- 11:00-11:30 **ESCG Opening Plenary:**
Session Chair: TBC
Speaker: Pedro Nunes, Diamond Light Source
Title: TBC
- 11:30-17:15 **ESCG Research Sessions, Flash Poster Presentations, and AGM**
- 17:15-17:45 **Parkin Lecture**
Session Chair: TBC
Speaker: Struan Simpson, University of Warwick

MAIN MEETING

- Room: PX001, Exhibition Centre**
- 18:00-18:45 **Lonsdale Lecture**
Session Chair: TBC
Speaker: Briony Yorke, University of Leeds
- 18:45-19:00 **Exhibitor Talks**
- 19:00-21:00 **Poster Session with Dinner and Wine**



Tuesday 23rd March 2027

BCA 2027 MAIN MEETING PROGRAMME

- 08:30-08:15 **CCG Plenary**
Room: PX001, Exhibition Centre
Session Chair: TBC
Speaker: Jon Steed, University of Durham
Supramolecular Gels – Tangled Soft Materials
- 09:15-10:00 **IG Plenary**
Room: PX001, Exhibition Centre
Session Chair: TBC
Speaker: Richard Walton, University of Warwick
Title: TBC
- 10:30-17:00 **Parallel Sessions & Interest Group AGMs**
- 17:10-17:55 **BSG Plenary**
Room: PX001, Exhibition Centre
Session Chair: TBC
Speaker: Andrew Lovering, University of Birmingham
Title: TBC
- 18:00-19:00 **BCA Annual General Meeting**
- 19:30-00:00 **Conference Dinner, Networking & Ceilidh**

Wednesday 24th March 2027

- 09:00-09:45 **PCG Plenary**
Room: PX001, Exhibition Centre
Session Chair: TBC
Speaker: Caroline Kirk, University of Edinburgh
Title: TBC
- 09:45-09:55 **Exhibitor Talks**
- 10:30-14:10 **Parallel Sessions**
- 14:40-15:25 **Bragg Prize Lecture**
Session Chair: TBC
Speaker: Announced on the day
- 15:35-16:45 **Early Career Prize Lectures**
Biological Structures Group Early Career Prize
Announced on the day
Chemical Crystallography Group Prize for Younger Scientists
Announced on the day
- 17:05-17:20 **Poster prizes and close**
Room: PX001, Exhibition Centre

CLOSE OF CONFERENCE

2026 Spring Meeting reports

ESCG sessions

ESCG Opening plenary

The BCA spring meeting 2026 was once again kicked off with the ESCG satellite meeting held in the familiar territory of the University of Leeds and proved to be an incredibly productive, lively, and informative affair, as usual. The opening plenary session was chaired by **Sam Lewis** (EuroXFEL). Our plenary speaker for this year was the wonderful **Ines Collings** (Natural History Museum) who gave a fascinating talk entitled “Salt hydrates on icy moons: structures and properties under extreme conditions.” Ines explained that her research is based on interpreting data from space missions to the outer planets using ground-based laboratory results from high pressure and low temperature studies of crystal growth from salt water. The speaker explained that the ‘snow line’ of the solar system lies somewhere between the planets Mars and Jupiter. Ines expertly detailed the conditions that exist on the hundreds of small moons that orbit Jupiter and Saturn, and how they can be investigated here on earth with the use of diamond anvil cells.

Several of these moons (e.g. Europa, Ganymede and Callisto) are thought to have liquid salt water oceans beneath their ice crusts and some have more ocean than the earth. These moons also contain silicate rock and volatile organics and, intriguingly, Enceladus, a moon of Saturn, is known to eject water plumes. The speaker summarised the history of space exploration to the outer planets, beginning with the Pioneer and Voyager missions of the 70s and 80s, through to Galileo, Cassini and Juno, as well as the contemporary missions Europa and Juice which are due to arrive at Jupiter in around 2030. Instruments carried by these probes include mass spectrometry, dust analysers, infra-red and UV, with the final interpretation of the results relying heavily on laboratory standards. In that vein, the speaker presented a detailed phase diagram of ice with temperature and pressure as variables and explained that her current research involves studies of the stability of solid-liquid interfaces, new salt hydrates and their potential role in vertical transport to the surface.

Crystallography is used to provide information on structure and phase, chemical composition, density and bulk modulus. Current work involves building a database of salt hydrate phases and structures with ongoing experimental studies, including the use of membrane driven diamond anvil cells with cryostatic cooling. The high temperature and pressure facility at the ESRF beamline ID15B allows work of this sort to be undertaken with μm scale samples. This has allowed the speaker to discover new phases of chloride salts including NaCl hydrates and intriguing layered structures in MgCl_2 ices which have anisotropic compressibility. This was attributed to staggering of the octahedra in one direction in the crystal structure. A general finding in these studies is that

hydration number increases with pressure. The speaker then described spectroscopic studies including near infra red work on high pressure salt hydrates using a piston-cylinder press followed by cryo-recovery to ambient pressure, as well as the assembly of a database of associated results for comparison with remote sensing experiments.

Ines also explained that her research had origins in the field of metal organic frameworks, and that she was first introduced to her current field while working with the Matter at Extremes group at the ESRF and supporting a user experiment. This provided an incredibly important reminder to members of the ESCG that we should always seek out new opportunities and engage with the community as you never know where it will lead you! The opening plenary truly set a wonderful tone for the rest of the meeting.

ESCG Session 1

This session was chaired by **Jake Hill** (Leeds) and began with a lecture by **Nicola Kelly** (Cambridge) entitled “Structure and magnetism in the distorted square lattice compounds $\text{M}'\text{-LnTaO}_4$.” Nicola’s talk began with an introduction to thermal barrier coatings which are used to protect the metal parts of gas turbines. Nicola explained that her main interest is actually in low temperature studies of this material which has been shown to adopt 3 phases M and M’ which are monoclinic (C2/c or I2/a and P2/c, respectively) and the T phase which is tetragonal ($I4_1/a$). The different phases arise from shearing of the lattice structure which can be brought about with different regimes of heating and compression. Having synthesised M’ phases across the lanthanide series (Tb, Dy, Er and Ho) Nicola characterised them by synchrotron PXRD, powder neutron diffraction and magnetic susceptibility experiments at both room-temperature and low temperatures (30 K and 1.5 K). Neutron analysis of larger single crystals was also undertaken. Nicola showed how the M’ phase adopts a stretched diamond lattice with two-dimensional square layers, giving rise to magnetic behaviour markedly different from that of the corresponding M phase with ferromagnetism and antiferromagnetism being reported. The magnetic unit cell turns out to be twice as big as the nuclear cell in two dimensions. The speaker ended by summarising the prospects for future low temperature work on other lanthanide compounds.

Alex Johnson (Newcastle) then spoke on “High-throughput crystallisation of polar biologically relevant small molecules” such as cyclic peptides. The standard ENaCt workflow had limited success but by switching to polar encapsulation oils such as PEGs and matching oil polarity to the solvent polarity, Alex was able to improve both crystal quality and hit rates, ultimately obtaining a crystallisation hit for cyclosporin A, opening the door to broader application of high-throughput methods in this difficult chemical space.

During his talk the speaker covered various techniques of high-throughput screening for crystallisation conditions for



Speakers and chairs from the ESCG plenary and first session. Left to right: Jake Hill (Leeds, chair), Eleanor Keil (Southampton), Alex Johnson (Newcastle), Nicola Kelly (Cambridge), Ines Collings (Natural History Museum) and Sam Lewis (EuroXFEL, chair).

organic compounds and mentioned how cyclic peptides (MW 1000 – 5000 Da) present additional challenges. Alex covered the ENaCt method of robotic crystallisation which has very low sample requirements and allows one to vary properties of the encapsulating oil such as its polarity [1]. The basis of this method is the slow partitioning of the solvent through the encapsulating layer of oil and into the atmosphere. Amino acids were tested before moving on to dipeptides using water and tertiary butanol as the solvent. This revealed that a polar “oil” such as PEG400 seems to improve the crystallisation outcome when water is the solvent. Polypropylene glycol (PPG), ethylene glycol and diethylene glycol as well as low molecular weight polyethylene glycols, such as PEG 400 and other compounds which vary in water solubility and liquidity, were successful as encapsulants.

The session concluded with **Eleanor Keil** (Southampton) who described the development of devices for *in situ* gas absorption experiments compatible with both X-ray and electron diffraction at home lab sources. After several design iterations to overcome issues with source compatibility, phi angle restrictions and gas leaks, Eleanor presented a new single-piece welded design and outlined validation using CALF-20, a zinc-based metal organic framework (MOF), as a model system for CO₂ capture.

Eleanor covered the T-bar gas cell, the elbow cell and the Microfit cell, as well as the welded gas cell and an especially manufactured cell for SCXRD. The latter allowed data collection on CALF-20 which is a zinc triazole and oxalate paddle wheel structure that was amine-functionalised. Studies using Platon Squeeze [2] suggests that CO₂ binds at a static pressure of 2 bars. Use of a gas chip manufactured by Hummingbird in electron diffraction studies of this system allowed measurement of CO₂ binding at 1.1 bar.

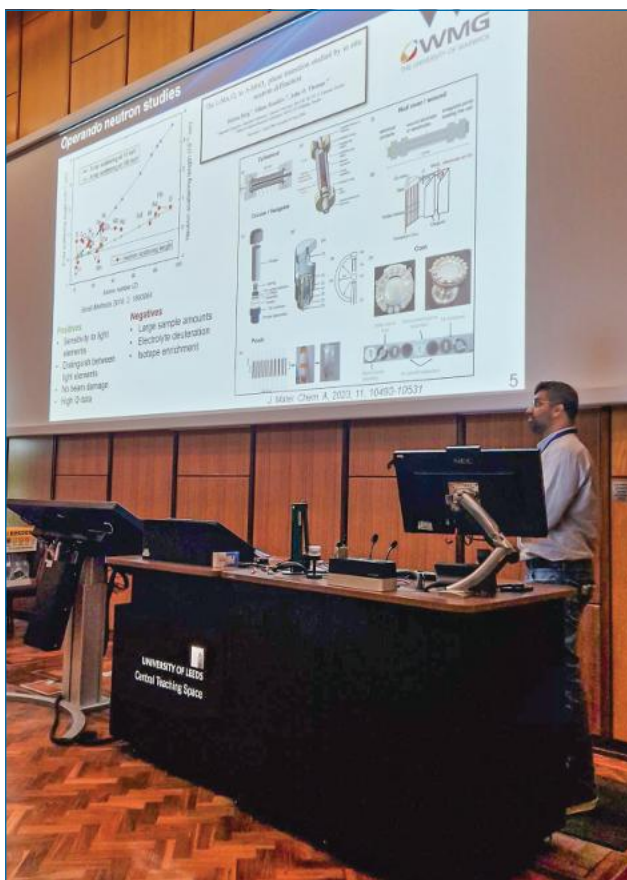
The session concluded with the flash poster presentations in which each speaker has one minute to outline the key findings of their work and encourage attendees to view their poster during the lunch break.

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ESCG Session 2

The second ESCG session was chaired by **Celine Beck** (Glasgow/DLS) and began with a talk from **Ashok Menon** (Warwick) who spoke on the subject of how “Cold neutrons enable *operando* neutron diffraction of unmodified, non-deuterated, non-isotope-enriched battery cells.” Ashok talked about how cold neutrons can be used as a powerful tool for *in operando* battery research. Studying battery devices with realistic geometries is both difficult and crucial for their development. Batteries also usually contain materials with large incoherent scattering or absorption cross sections such as H and Li, leading to noisier data and lower time resolution. Ashok outlined how both the long-wavelength neutron flux-profile and geometry of the WISH beamline at ISIS has enabled him to gather high quality neutron diffraction data of non-deuterated industrially relevant battery cell geometries and better understand their deterioration upon charge-discharge cycles.



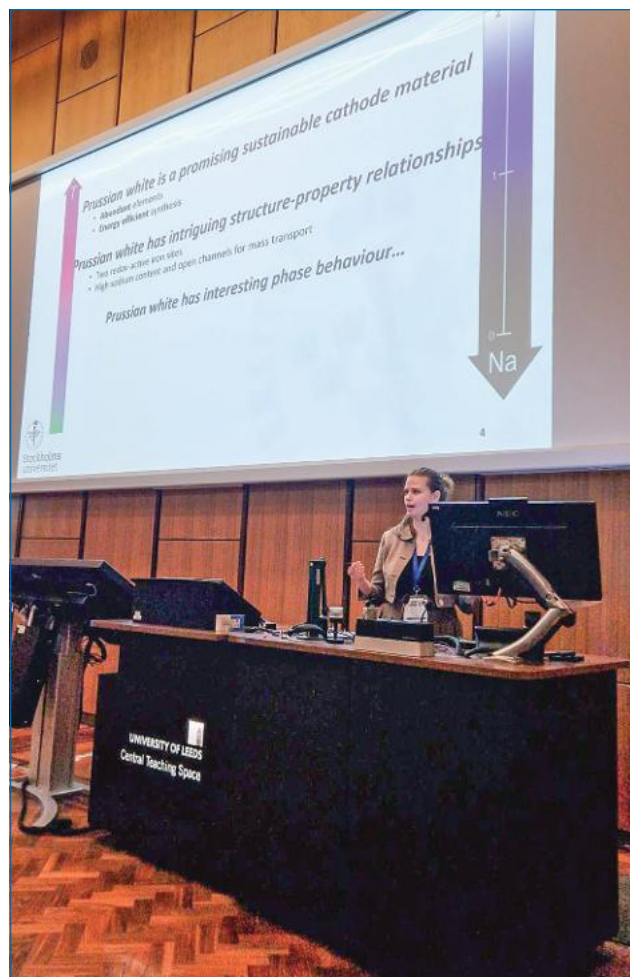
Ashok Menon (Warwick) speaking on *operando* neutron diffraction.

Ashok began by outlining the design of battery cells which consist of an anode, a separator and a cathode material, such as LiCoO_2 . The speaker mentioned how lithium ions move between the electrodes during battery operation and long-term electrochemical cycling alters the properties of the components. The speaker mentioned how batteries are, of course, kinetically stabilised systems rather than being thermodynamically stable and how X-rays induce damage in them. In contrast, neutrons provide a damage-free method of studying batteries and are more sensitive to the very light elements, such as H and Li, which have poor visibility with X-rays. The speaker emphasised the complexity of studying traditional batteries with their multi-phase solid, liquid and

gas components. Neutron diffraction methods usually require large sample sizes and deuterated or other isotope-enriched components. Ashok described the use of single- or multi-layer pouch cells as the preferred system of study and outlined how single pouch cells have been successfully studied by neutron scattering using the WISH time-of-flight cold neutron powder diffractometer at Target Station 2 of ISIS. This installation yielded better results than Polaris which generates thermal neutrons. The speaker described time- and voltage-dependent studies of fresh and aged cells which have been through 100 charge and discharge cycles.

Elina Elvelo (Stockholm) did an excellent job giving her first conference presentation entitled “Pressure response in Prussian White – multidimensional phase diagram of an important cathode material” covering both the temperature and pressure response of this compound. This material is a promising alternative to conventional battery materials which typically use less abundant elements. While Prussian White had been well studied at various temperatures, its pressure behaviour is much less well understood. Elina recounted how neutron powder diffraction enabled her to explore the P-T phase diagram, revealing a previously unknown superstructure at high pressures.

The speaker explained how batteries have been made from a range of materials including Li, Ni, Mn, Co and graphite. A sodium ion battery would have the advantage of avoiding the need to obtain rare materials, hence the interest in compounds such as Prussian White ($\text{Na}_2\text{Fe}[\text{Fe}(\text{CN})_6] \cdot n\text{H}_2\text{O}$).



Elina Elvelo (Stockholm) speaking on the “Pressure response in Prussian White.”

The speaker explained how the space group of this compound changes with increasing temperature whereas the dehydrated form does not exhibit this effect. It has two redox iron sites, a high sodium content and open channels which could facilitate sodium ion diffusion during charging and discharging.

Elina described high pressure studies undertaken at ISIS-Polaris which are aimed at taking advantage of the enhanced contrast of the elements C and N, as well as the lower beam damage encountered when using neutrons rather than X-rays. The speaker mentioned the use of diamond anvil cell X-ray studies and also described work at ISIS-Pearl, a powder diffractometer dedicated to the measurement of crystal structure parameters under high pressures, coupled with high or low temperature conditions. Use of these instruments demonstrated that Prussian White undergoes multiple phase changes as the pressure increases, specifically the space group changes from $R\bar{3}$ to $P2_1/n$ above 0.5 GPa but then it reverts to $R\bar{3}$ above 3.2 GPa. The speaker described new evidence for superlattice formation at high pressure.

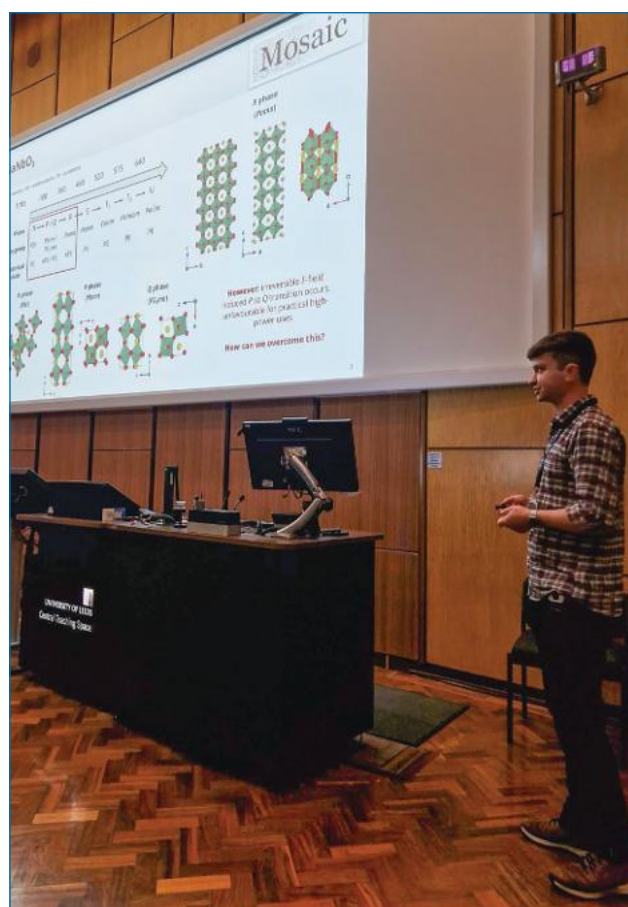
Next up, **Debashish Das** (Cardiff) presented his research on “Tuning the properties of spin crossover (SCO) complexes by counteranions and lattice solvents” by introducing them into the lattice. Spin crossover materials have many applications, including sensors, molecular switches, or memory devices. Debashish outlined how both single crystal and powder XRD were used to study these transitions, which were induced by varying temperature or

light exposure. These studies revealed how the transitional behaviour of transition metal complexes can be fine-tuned by introducing counter ions or solvents into the structure.

The speaker covered octahedral transition metal complexes with d4 – d7 electronic configurations which exhibit switching from low to high spin that is inducible with temperature, pressure, light and a magnetic field. Debashish described the synthesis and crystallisation studies of the bisphenol SCO compound $[\text{Fe}(\text{bpz})_2](\text{ClO}_4)_2$ with different counterions and solvents. Powder and single crystal diffraction studies have shown this compound to change reversibly from a black $P2_12_12_1$ crystal form to a yellow $P4_2/c$ form upon heating. Using light to induce the low-to-high-spin state transition gives a long lived state. The speaker described studies in which the crystallisation solvent was varied along with serial femtosecond X-ray studies at DLS and SwissFEL using the pink beam on microcrystals.



Debashish Das (Cardiff) presenting on spin crossover complexes.



Ben Tragheim (Sheffield) speaking on quasi-linear dielectric systems.

Ben Tragheim (Sheffield) gave an excellent overview of his research looking into structure-property relationships of ceramic capacitors. Ben's lecture was entitled “Resolving local structural influences in quasi-linear dielectric, NaNbO_3 -based systems for high power, energy storage applications.” These quasi-elastic dielectric materials are being studied for their high power energy storage applications. He discussed how his work used X-ray total scattering, Rietveld refinement and pairwise distribution functions (PDF) to investigate both average and local structure of these materials. For future work, he intends to perform these measurements at variable temperature and using neutrons, which will be complemented by reverse Monte Carlo modelling.

Ben described how the electric properties of sodium niobate vary with temperature, changing from being ferroelectric at low temperatures, becoming antiferroelectric as the temperature is increased and finally paraelectric above about 640 °C. These effects were attributed to octahedral tilts in the structure and it was reported that tantalum doping creates ceramic materials with good energy density properties.



Chair of the second ESCG session Celine Beck (Glasgow/DLS) with Debashish Das (Cardiff), Elina Elvelo (Stockholm), Ashok Menon (Warwick) and Ben Tragheim (Sheffield).

ESCG Session 3

Following coffee and posters, **Emma Hawking** (CCDC) chaired the next ESCG session which began with a lecture by **Celine Beck** (Glasgow/DLS) entitled “High resolution powder diffraction under pressure: studying transient breathing states of MIL-53(Cr) using ISODISTORT.” Celine covered MOFs and the barocaloric effect which they exhibit, focussing on MIL-53(Cr) which absorbs water, even from the atmosphere. Barocalorics are promising candidates for solid state refrigerants which should be more environmentally sustainable than the fluids currently used for this purpose. The speaker used a liquid pressure cell at I11 (DLS) to study the transition in MIL-53(Cr) using PXRD and this yielded a high temperature structure into which water molecules were being built. The speaker described ongoing work using Rietveld refinement of the low temperature structure, analysis of which was being undertaken using rigid unit modes (RUM) to find rocking modes that fitted the data. The latter work yielded two plausible structures.

Next up, **Džonatans Melgalvis** (Latvia) spoke on the subject of “GraPhAI: graph neural networks for solving centrosymmetric crystal structures” and began with the impressive observation that whilst direct methods approach fails above 1.6 Å resolution, AI is capable of solving small centrosymmetric structures using 2 Å data. The speaker described how the reciprocal lattice which is traditionally described with indices h , k and l can be represented more efficiently by a graph convolution network and how the reflections can then be assigned random phases. Since the CSD does not currently possess a sufficient number of structures to train the algorithm, something like 150 – 180 million theoretical structures were used for training the neural network. Iterative refinement of the phases deduced by the network improves the results. Džonatans described how the AI approach is more successful at solving structures with a heavy atom ($Z \geq 19$). Indeed, something like 80 % of centrosymmetric structures with a suitable heavy atom can

be solved. The speaker finished by summarising the key features of the approach with its new efficient representation of the diffraction data and ability to solve centrosymmetric X-ray and ED structures from the CSD and COD. The success rate with purely organic structures is approximately 40 % suggesting that this forms a challenging avenue for future research.

Giovanna Barrionuevo Martins (Newcastle) then spoke on “Synthesis-driven structural elucidation of challenging carbonyl containing compounds via functionalised crystallisation.” The speaker outlined approaches for derivatising molecules as a means of crystallising them, as well as the ENaCt (encapsulated nanodroplet crystallisation) approach which involves screening in a 96 well plate format. Due to the use of solvents in the crystallisation of organic compounds, the plates need to be made of glass. Giovanna then described an approach that has been successful with compounds which still refuse to crystallise and this involves the use of reagents such as 2,4-dinitrophenylhydrazine (Brady’s reagent) and the safer compound 4-bromo-2-nitrophenylhydrazine (Martins reagent). These can be used as *in situ* covalent crystallisation reagents and often yield crystals within an hour. Aldehydes were found to react well but ketones need acid in order to react. The speaker described a huge volume of work on crystallisation and structure analysis of aliphatics, heteroaromatics, α/β -unsaturated and cyclic compounds, including steroids and non-steroidal anti-inflammatory drugs (NSAIDs) as well as 3-formyl rifampicin. The latter is an intermediate in the synthesis of rifampicin, an antibiotic which is used as a first-line treatment for TB. Many of these compounds are large greasy molecules which would normally be difficult to crystallise.

ESCG Session 4

After the ESCG AGM and coffee break, there were two more contributed talks beginning with **Joseph Burke** (Imperial) who spoke on “Predicting crystal structures of porous materials using coarse-grained modelling.” The speaker explained how the established crystal structure prediction techniques work by breaking the workflow into three stages, namely sampling, screening and ranking. These techniques have shown excellent accuracy but are highly computationally expensive and scale poorly when screening a large array of molecules and multicomponent systems. Joseph then described coarse grained modelling of systems, such as the extended benzimidazole compound T2, by treating charge and polarity of groups as patches on spheres representing the atoms to achieve greater computational efficiency in the prediction of porous organic phases. The coarse graining allows greater time- and length-scales to be studied and is frequently applied to biological systems. The speaker’s project involves the use of Monte Carlo methods to simulate cooling of the T2 system and this allowed the porous phase to be predicted. The speaker described an automated way of finding phases in the simulation with the aim being to work out the driving forces for the formation of different polymorphs.

Last but not least, the final speaker in this session was **Alexandra Longcake** (Liverpool) who described “Establishing an advanced diffraction facility as an early career researcher.” Alexandra described the setting up of



Dzonatans Miks Melgalvis (Latvia), **Ben Tragheim** (Sheffield), **Joseph Burke** (Imperial), **Alexandra Longcake** (Liverpool), **Emma Hawking** (CCDC, ESCG sessions 3 and 4 chair), **Celine Beck** (Glasgow/DLS, ESCG session 2 chair), **Georgia Orton** (Birmingham) and **Giovanna Barrionuevo Martins** (Newcastle).

the Materials Innovation Factory in Liverpool with a Synergy 3DED and a SC-XRD system in the form of a Rigaku Synergy Dual-Wavelength (DW) Flow instrument. The equipment was installed in late 2025 effort as an EPSRC-supported open access translational research facility. The processes of equipment installation, testing procedures and establishing laboratory infrastructure were covered with the aim of encouraging other early-stage crystallographers who are interested in moving into facility management roles in the future.

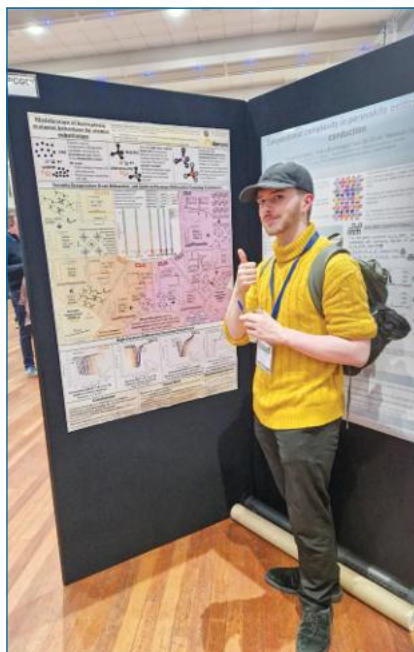
Reports on the Parkin and Lonsdale Prize Lectures (which followed this session) appeared in the previous issue of Crystallography News.

Celine Beck, Glasgow
Jon Cooper, UCL
Jake Hill, Leeds
Evie Ladbrook, Warwick
Sam Lewis, EuroXFEL
Mark Montgomery, Syngenta
Natalie Pridmore, Durham



The 2026 Parkin Prize Lecturer **Georgia Orton** (Birmingham) with **Ben Tragheim** (Sheffield, chair).

An assessment of the silver plaques on the oak stand of the BSG David Blow Poster Prize (a Blue John crystal, which is partly visible at the top) suggested that these are almost up-to-date, so thank you to previous winners! This year the prize was awarded to **Charlotte Hunter** (Durham) for the poster "Studying biomolecular interactions with the Dianthus screening platform." A full report on the BSG sessions at the 2026 Spring Meeting appeared in the last issue.



One of the Spring Meeting poster sessions and the winner of the ESCG poster prize **David Coventry** (Edinburgh). Photos courtesy of **Celine Beck** (Glasgow).

IG Plenary

The IG Plenary was chaired by **Tony Bell** (Sheffield Hallam) and was given by **John Helliwell** (Manchester/Daresbury) with the title of “Open Science and Industry.”

John introduced the concept of Open Science and the international organisations such as the UN and UNESCO which support this concept. Science is considered as a Public Good, but attitudes to Open Science are different around the world. Europe is pushing open science to society at large as the default. However, China by contrast is emphasising legal obligations so “open wherever possible, closed where necessary.” In countries like Canada, Australia, and New Zealand, UNESCO have dropped “open by default” and adopted “as open as possible, as closed as necessary” for the protection of indigenous peoples’ traditional knowledge, data sovereignty, and intellectual property rights.

Open Science in an industrial scientific capacity involves co-operation between different industrial organisations. The IMCA-CAT beamline at the Advanced Photon Source has different pharmaceutical companies collaborating in structural biology. Similarly, the DLS “COVID moonshot” was used to study protein structures of SARS-CoV-2. These represent Open Science collaborations. New drug discoveries have had the same innovation model with open foundational knowledge leading ultimately to proprietary medicinal chemistry and finally regulated commercial product.

Prof. Helliwell and I both used to work at the Daresbury SRS. The Daresbury DARTS service introduced synchrotron radiation to industrial customers. Synchrotron sources do not keep non-commercial data hidden indefinitely. The ESRF releases non-proprietary data to the public three years after data collection.

The UKRI is dedicated to open research with the aim of achieving openness, verification and reproducibility as well as improving public value, research integrity, reuse and innovation. The overriding facts are that companies compete commercially and consumers have to have choice, coupled with the regulatory and judicial protection of all involved. With this in mind, the speaker looked at the question of who solves societal challenges such as climate change, disease and future industry / employment, raising the question of when industry might ever adopt open science. In such cases, a fully open collaborative approach, supported by UNESCO and WHO, could accelerate breakthroughs of global importance. The speaker stated how the UN Science Advisory Board promotes sustainable development goals with the UN Industrial Development Organisation stating that “inclusive growth decoupled from emissions” is one of its principal aims.

The speaker emphasised the importance of conducting crystallography experiments at body temperature (37 °C or 310 K) to yield patient-relevant structures. John described how the IUCr is a pioneer of peer review and data deposition as well as encouraging authors to make their original measurements available in raw data archives. The speaker described the fairification of data which has the aim of making experimental measurements findable, accessible, interoperable and re-usable (FAIR) with globally unique and persistent identifiers and rich metadata. John noted that these aims are not restricted to open data and concluded by mentioning that we need to develop the Global South, with efforts to establish the African Light Source (<https://www.africanlightsource.org/>) being an exemplary model of work in that direction.

Tony Bell, Sheffield Hallam



*Chair of the IG Plenary Lecture **Tony Bell** (Sheffield Hallam) with the invited Plenary speaker **John Helliwell** (Manchester/Daresbury).*

CCG/ESCG/IG Crystal Formation

The CCG/ESCG/IG “Crystal Formation” session was jointly chaired by **Andrea Laybourn** (Leeds) and **Sam Lewis** (EuroXFEL) and comprised 4 talks covering crystallisation, crystal prediction and crystal engineering. The session began with a keynote from **Sarah (Sally) L Price** (UCL) entitled “What is needed for the ideal organic crystal structure prediction code?” The talk covered work conducted under the Bringing Experiment and Simulation Together in Crystal Structure Prediction (BEST-CSP) COST Action (CA22107), involving 41 authors. Sally described *ab initio* computational approaches used to determine thermodynamically stable polymorphs of the drug sulfamerazine (4-amino-N-(4-methylpyrimidin-2-yl) benzenesulfonamide, SMZ), both with and without the methyl group. Sally outlined a range of advantages and limitations of CSP approaches, including the treatment of thermal effects, lattice energies, dispersion corrections, as well as intramolecular and lattice vibrational modes. Sally then went on to discuss whether crystal structure prediction (CSP) could be used to design optimal crystallisation experiments, commenting on the conditions required to form and/or convert between SMZ polymorphs, such as slurring, seeding, temperature and solvent choice, as well as the challenges associated with experimental characterisation. In particular, Sally highlighted the need for high-quality powder X-ray diffraction (PXRD) data to reliably distinguish between Forms I and V. Finally, Sally noted that a robust, universal CSP approach should incorporate accurate lattice energies and thermodynamic data, including both polymorph and property prediction. Ultimately, predicting crystallisation kinetics remains particularly challenging, both computationally and experimentally; however, significant progress has been made in the field.

Next, **Elizabeth Galtry** (EuroXFEL) gave an invited talk on the topic “Dynamic sample environments for *in situ* studies

of crystallisation.” Beth began by providing an overview of crystallisation environments, including segmented flow and acoustic levitation, with reference toward their use for crystallisation methods such as cooling, reactive and antisolvent approaches. Beth then went on to showcase segmented-flow crystallisers equipped with *in situ* powder X-ray diffraction (PXRD), developed through collaboration between the University of Nottingham and Beamline I11 at DLS. This was exemplified by a study on the slurring of polymorphic *o*-aminobenzoic acid and the investigation of solvent-mediated phase transitions. Beth concluded the talk by highlighting acoustic levitation as a “contactless” sample environment, available at both Beamline I19 (where potential case studies include evaporative, reactive and antisolvent crystallisation of nanolitre-scale samples with *in situ* PXRD) and the Materials Imaging and Dynamics (MID) instrument at the European XFEL, which is equipped with PXRD, SAXS and XPCS for studying microlitre droplets over 10 – 40 minute timescales.

Tony Keene (University College Dublin) gave an enthusiastic third talk, entitled “Hydrogen bond network descriptor for the oxalic acid / oxalate system” which described an investigation into oxalic acid / oxalate co-crystals and a notation method he has developed from scratch to describe these systems in a detailed and systematic way. Tony noted that, despite hydrogen oxalates and oxalic acid being among the oldest crystal structures to have been solved, a dedicated descriptor for these structures is not yet reported. The descriptor developed and presented by Tony considers a range of structural features, including O-H ladders and chains; *cis* and *trans* isomers; flat and perpendicular geometries; bound and free hydrogen atoms and disorder; C-C bond orientations; the number of subunits; and Rosette and Frieze groups. Tony’s presentation highlighted the importance of robust descriptors in ensuring accurate, consistent and unambiguous identification and comparison of the many known structures within oxalic acid / oxalate crystal systems.



Speakers and chairs in the CCG/ESCG/IG “Crystal Formation” session. **Sam Lewis** (EuroXFEL, chair), **Tony Keene** (University College Dublin), **Elizabeth Galtry** (EuroXFEL), **Sally Price** (UCL), **Georgia Orton** (Birmingham) and **Andrea Laybourn** (Leeds, chair).

The session concluded with a talk from **Georgia Orton** (Birmingham) entitled “Programmed assembly-disassembly: harnessing solid-state organisation and reactivity to access previously unreachable molecules”. The talk covered a range of aspects related to order and disorder, as well as flexibility and reactivity in the solid state. It showcased the use of ozone-induced disassembly of aldehydes and ketones to access materials that are not obtainable by other means. A key finding was that solvents template the crystal structures, forming layers of interdigitated cages. Upon irradiation at 365 nm, the materials underwent a solid-state [2+2] cycloaddition reaction to produce a cyclobutene moiety. This transformation required a propeller-like rotation over several hours, which was tracked through the determination of intermediate crystal structures and the mapping of electron density, revealing arm rotation within the crystal. Final disassembly was achieved by the addition of hydrochloric acid, which released the target materials by breaking the metal-ligand bonds. Georgia’s talk perfectly exemplified the importance in rational synthetic approaches to achieve optimal structural properties for these kinds of studies.

Overall, the session exemplified the importance of collaboration – from large-scale efforts involving 41 authors across the EU, to partnerships with researchers at international facilities such as the EuroXFEL and Diamond Light Source (DLS), as well as hosting family placements and Master’s students. The session also highlighted the strong complementarity between experimental and computational approaches and demonstrated the value of drawing inspiration from the literature, alongside the need for accuracy, reproducibility, and effective communication in scientific research.

Andrea Laybourn, Leeds
Sam Lewis, EuroXFEL

PCG / IG PDF workshop

The joint PCG / IG workshop on PDF analysis was chaired by **Tony Bell** (Sheffield Hallam) and began with a lecture by **David Keen** (ISIS Neutron and Muon Source) entitled “Rosalind Franklin used the pair distribution function in her research; how – and why – you should too.”

The pair distribution function (PDF) – total scattering technique is a combination of Bragg and diffuse scattering. I first became aware of this at the 2006 European Powder Diffraction conference in Geneva, where I saw a fascinating presentation by Simon Billinge. I did a little bit of work using this technique in the next couple of years but nothing since then. In my ignorance I assumed that this new (to me) technique was developed in the late 1990s or turn of the century.

However, Dr. Keen’s presentation showed that this method has a much longer history. Friedrich (Laue’s student) first noticed diffuse scattering in 1913! The PDF technique was first discussed by Zernike and Prins (1926) and PDF measurements on liquid mercury were done by Debye and Menke (1930). The first solid crystal PDF measurements were done by Warren and Gingrich (1934). Rosalind Franklin, before she became posthumously famous for her work on DNA, did some PDF work on nanocarbons in 1950. She did the Fourier calculations by hand! How many of us could do that today?

The technique became more practical from the late 1980s and early 1990s, with the advent of spallation neutron



Chair of the PCG / IG PDF workshop Tony Bell (Sheffield Hallam) with the speaker Dave Keen (ISIS).

sources and high energy synchrotrons, as data could be collected further out into Q space. Also, the development of faster computers speeded up the Fourier calculations.

Why use this technique? The local structure can be different to the average structure, disorder can be key to a material’s properties. If your samples are poorly or nanocrystalline, you can say more than the sample is amorphous. However, care must be taken to collect good data, as far out in Q space as possible. Additionally, care must be taken to correct your data before Fourier transformation, corrections must be taken to account from things like scatter from sample holders. Be aware that the mathematics involved in these data analyses are rather complicated.

Dr. Keen illustrated the presentation with some of his research over the years, mostly using neutrons although there was also some X-ray work. He ended the presentation with talk of future developments. It is now possible to do PDF experiments using laboratory X-ray sources which have silver anode X-ray tubes. In the future, time-resolved PDF measurements will be possible at FEL sources.

Tony Bell, Sheffield Hallam

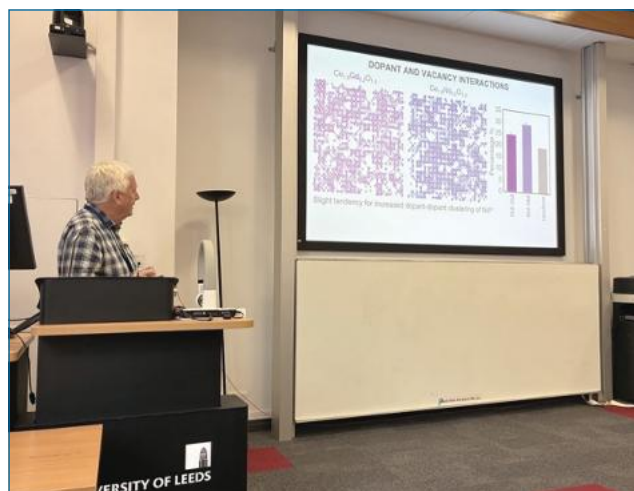
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PCG Energy Materials

The Energy Materials session of the Physical Crystallography Group at the BCA Spring Meeting 2026 was chaired by **Ashok Menon** (Warwick) and contained 4 talks. The session began with a keynote lecture entitled “Exploring structure-property relationships in organic-inorganic metal halides for optoelectronic applications” which was given by **Julia Payne** (St Andrews). The speaker presented their work on tuning organic-inorganic metal halides via halide substitution and intercalation. Julia discussed the structural and optoelectronic properties of layered bismuth halide perovskite systems, showing how substitution of iodide by bromide in phenylenediammonium bismuth iodides allows systematic tuning of the band gap. Crystallographic methods were combined with computational screening to identify the most promising layered perovskite candidates across a range of organic-inorganic and inorganic bismuth halide compositions. Proof-of-concept scale-up of devices was also demonstrated.

Next, **Stephen Hull** (ISIS Neutron and Muon Source) spoke on the subject of “The ISIS facility – developments for neutron powder diffraction studies of energy materials.” The speaker gave an overview of the complementary roles of X-rays and neutrons in the study of energy materials, before focusing on cerium-based solid electrolytes as alternatives to commercial yttria-stabilised zirconia (YSZ) for solid oxide fuel cell applications. Stephen presented *in situ* heating studies of gadolinium-doped CeO_2 under controlled oxygen atmospheres, analysed using total scattering data and reverse Monte Carlo (RMC) modelling with the ^{160}Gd isotope. A central theme was the nature of vacancy-vacancy correlations within the fluorite structure, with complementary computational work demonstrating that substitution of the larger Nd^{3+} ion reduces vacancy-vacancy repulsion, which is a critical insight relevant to optimising oxide ion conductivity in these materials.



Stephen Hull (ISIS Neutron and Muon Source) speaking on developments for neutron powder diffraction studies of energy materials.

Erin Carroll (Sheffield) then presented work on NaNbO_3 as a dielectric and capacitor material, showing how doping with Nd suppresses the P-to-Q structural phase transformation. Erin's talk was entitled “The importance of neutron diffraction for understanding structure – property relationships in A-site deficient NaNbO_3 .” *In situ* neutron diffraction proved particularly powerful in this study, revealing an orthorhombic R phase characterised by octahedral tilting and Nb displacements that are not apparent from X-ray diffraction alone, which reflects only the average cubic P phase structure. Relative permittivity measurements were used to characterise the resulting electrical properties.



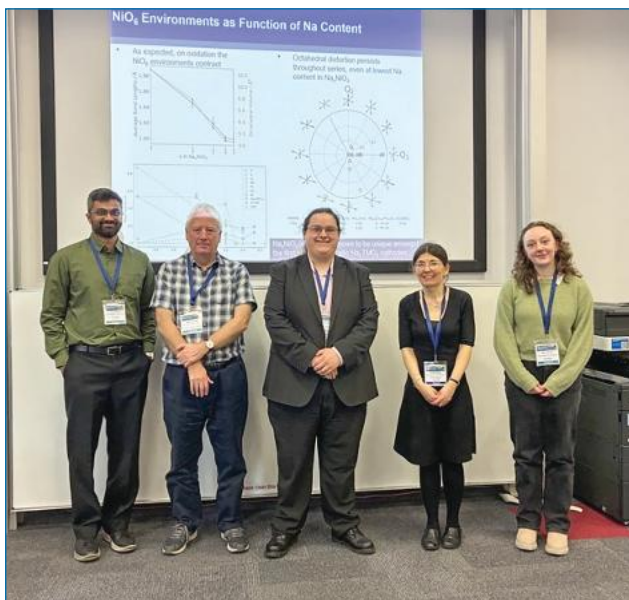
Erin Carroll (Sheffield) speaking on neutron diffraction of sodium niobate.

The session concluded with a talk entitled “Sodium vacancy, nickel charge and orbital order: structural evolution in electrochemically desodiated Na_xNiO_2 ” given by **James Steele** (Cambridge). The speaker focused on NaNiO_2 as a sodium-ion battery cathode material. James highlighted the practical advantages of sodium over lithium systems and presented a make-measure-model approach to understanding the relationship between structural and electrochemical properties. A notable finding was that NaNiO_2 retains some degree of octahedral distortion throughout cycling, distinguishing it from analogous layered oxide systems and offering structural insight into its electrochemical behaviour.



James Steele (Cambridge) speaking on NaNiO_2 as a sodium-ion battery cathode material.

Ashok Menon, Warwick



Chair of the PCG Energy Materials session **Ashok Menon** (Warwick) with speakers **Stephen Hull** (ISIS), **James Steele** (Cambridge), **Julia Payne** (St Andrews) and **Erin Carroll** (Sheffield).

PCG Quantum Materials

This session was dedicated to materials where quantum effects, in a general sense, are relevant for the behaviour. This can include various forms of magnetism, spin-orbit coupling and unusual electronic effects.

The PCG Quantum Materials session was chaired by **Nicola Kelly** (Cambridge) and kicked off with a keynote lecture from **Lucy Clark** (Birmingham) on “Tackling synthesis-dependent structure and properties in quantum materials.” The speaker introduced us to the different degrees of freedom important to quantum materials and then described the fascinating differences in physical properties between her samples of ZnV_2O_4 synthesised by conventional furnace heating vs rapid microwave-assisted heating. The latter route produces highly crystalline samples with highly reproducible structural and magnetic behaviour, placing ZnV_2O_4 as strong candidate for a quantum spin liquid state.

The next talk was an invited contribution from **Struan Simpson** (Warwick) on “Leveraging rigid-unit modes to design new magnetoelectric multiferroics.” Rigid-unit modes or RUMs are low-energy vibrational modes capable of driving phase transitions in crystals: readers may already know about these in the form of octahedral tilts in perovskites. Struan showed us his results on the new material $\text{CeBaMn}_2\text{O}_6$, featuring both neutron and electron diffraction, and an interesting comparative study of structure-magnetism relationships in 4H-SrMnO_3 and BaMnO_3 hexagonal perovskites.

Ashley Brennan (St Andrews) spoke about “Unconventional magnetism and residual entropy in $\text{La}_7\text{Os}_3\text{O}_{18}$.” Ashley has synthesised – with considerable experimental effort – this new layered osmate which has strong spin-orbit coupling thanks to the 5d Os^{5+} cation. Like its 4d relative $\text{La}_7\text{Ru}_3\text{O}_{18}$, $\text{La}_7\text{Os}_3\text{O}_{18}$ has a complex stacked crystal structure made of four different layers, and a similarly complex magnetic phase diagram with antiferromagnetic ordering at 43 K and

additional short-range ordering at 14 K. Detailed physical property measurements revealed that the low-temperature magnetism in this compound is non-trivial and we can look forward to learning more about it from upcoming neutron diffraction experiments.

The final talk of the session was from **Matthias Gutmann** (ISIS) on “Deciphering the sequence of phase transitions in KNiCl_3 .” This compound shows an intriguing series of structural phase transitions near and above room temperature, including both polar and non-polar structures. This also affects the electronic properties: there is significant variation in the dielectric constant, correlated with the structures. Matthias has solved several of these structures from single-crystal data. He also showed some magnetic data from neutron experiments on the related CsNiCl_3 and finished with a persuasive advert for proposals to use the SXD instrument (a single crystal time-of-flight Laue diffractometer at ISIS), where new detectors offer a count rate 4 times higher than that previously achieved.

Nicola Kelly, Cambridge



Matthias Gutmann (ISIS), **Struan Simpson** (Warwick), **Ashley Brennan** (St Andrews) and **Lucy Clark** (Birmingham) spoke in the PCG session on Quantum Materials.

PCG / CCG Functional Materials

The joint PCG / CCG Functional Materials session was chaired by **Ines Collings** (Natural History Museum) and began with the keynote presentation from **Anuradha Pallipurath** (Leeds) who presented a talk entitled “Colour-Changing Materials.” Anuradha detailed the various systems her varied career has touched on, moving from PhD work in medieval pigments via co-crystallisation studies of lawstone on to Pt pincer complexes which undergo stark colour changes when exposed to flowing dry nitrogen gas and to methanol. These colour-changes arise from absorption of solvent molecules into channels within the crystal structure which subtly alters the overlap of Pt centres, thereby altering the HOCO-LUCO energy gap and the colour. Anuradha’s further work covered thermochromic molecular complexes and purity assessment of iron sulphate feedstocks through the colour changes that occur with oxidation or the presence of impurities.

The second talk was given by **Charles McMonagle** (Swiss Norwegian Beamlines at ESRF) and was entitled

“Barocalorics and new tools for variable temperature and pressure *in situ* diffraction”. Charles outlined the case for barocalorics as refrigerants which avoid the environmental impacts of the gases which are currently used in the bulk of applications, and through his talk demonstrated the capabilities of the sapphire pressure cell used in the work undertaken at the ESRF. Samples are immersed in a liquid hydrostatic medium in the cell, and temperature-pressure phase space up to 1000 bar may be mapped automatically, additionally having the capacity to do Raman spectroscopy at the same time. Charles’ talk focused on the barocaloric material ammonium iodide, for which he presented an expanded phase diagram. He also explained that the high pressure diffraction techniques in his talk are complementary to the differential scanning calorimetry (DSC) measurements more typically used to study these systems, with the former delivering information about volumetric entropy change and the latter total entropy change.

Next was **Javier Castells-Gil** (Birmingham) who gave a talk describing how the “Interplay of local and long-range structure underpins electrochemical behaviour in disordered rock salt LiNiO_2 .” Nickelates present a possible alternative to cobaltates as cathode materials, and the disordered rock salt polymorph has advantages in capacity and energy density over the layered oxide structure, and it was these differences in the properties which Javier sought to understand. He used ball milling to synthesise the disordered rock salt, this being a metastable state while the layered oxide is the thermodynamic product. PDF analysis revealed additional structure in the data which could not be captured in the simple disordered cubic model, so Javier looked to big box modelling. He used reverse Monte Carlo simulations with 14,000 atoms, finding some clustering of the Ni and Li which impacts the percolation network and therefore the charge-discharge properties. He concluded with *in operando* PDF studies to track the evolution of these features during the charge-discharge cycle.

The final talk of the session was provided by **Rebecca Clulow** (Uppsala) and was entitled “Oxygen vacancy formation and stability of Al_2TiO_5 .” Al_2TiO_5 is an industrially relevant compound with uses based on its high thermal shock resistance, negative or low thermal expansion, and self-healing of cracks at elevated temperatures. However it also suffers from a complex phase diagram and instability over a wide temperature range. The material is

typically synthesised above 1300 °C, but is unstable and decomposes between 800 and 1300 °C. To synthesise bulk samples for neutron measurements, Rebecca and her co-workers explored different routes including ball milling, sol-gel synthesis, and water-quenching, eventually finding that standard solid state methods with quenching provided the best results with regard to phase purity. Neutron diffraction data from Al_2TiO_5 revealed oxygen vacancies, with these increasing in Fe doped samples. High temperature neutron experiments to track decomposition at 900 °C revealed the importance of these vacancies to the decomposition process.

Lewis Owen, Sheffield
Ben Tragheim, Sheffield

PCG Open Session

The PCG Open Session was chaired by **Ben Tragheim** (Sheffield) and started with a talk from **Emily Meekel** (Birmingham) entitled “Probing local polyhedral tilts and stability in a series of mixed-metal van der Waals frameworks.” Emily gave an incredibly interesting talk exploring the local structure of this novel class of materials, presenting elegant results from single crystal X-ray diffuse scattering and 3D- Δ PDF techniques, and describing how the interplay between the choice of metal cation and the disorder exhibited in the frameworks affects the material’s overall behaviour.

Following on from this, **Simon Cassidy** (Oxford) piqued the audience’s curiosity with his talk entitled “The curious crystal structure of cobalt.” Simon effectively illustrated that for such a chemically simple material such as cobalt, the crystals exhibit a complex structure composed of stacking faults between cubic and hexagonal close packed layers. Simon then demonstrated a rigorous method of how the stacking fault behaviour may be characterised through a grid search Rietveld refinement procedure of different models against synchrotron powder X-ray diffraction data.

Next, **Cameron Wilson** (Warwick) gave an engaging talk entitled “From molecular insight to hydrostatic performance: simulation-guided discovery of improved pressure-transmitting fluids.” Cameron highlighted key results that a change from the standardly used 4:1 methanol:ethanol



Charles McMonagle (ESRF), Javier Castells-Gil (Birmingham), Anuradha Pallipurath (Leeds), Rebecca Clulow (Uppsala) and Ines Collings (Natural History Museum, chair).



Chair of the PCG Open Session Ben Tragheim (Sheffield) with Michael Wharmby (Bruker), Cameron Wilson (Warwick), Simon Cassidy (Oxford) and Emily Meekel (Birmingham).

ratio to 9:1 for pressure-transmitting media (PTM) in high-pressure experiments gives an increase in the hydrostatic limit obtained. The reasoning for this was explored through molecular dynamics simulations and promising novel candidates for PTMs discovered using machine learning techniques were explored as well.

Finally, **Michael Wharmby** (Bruker) gave a fascinating talk on “Understanding structural complexity in M-CAU-64” covering this series of MOF materials which are important in green chemistry contexts. Michael demonstrated the range of different Bruker-based instrumentation (single crystal and powder diffractometers) to characterise the complex structural responses of these MOFs under variable temperature and atmospheric conditions.

It was a great joy to chair the PCG Open Session, and to hear such a wide range of intriguing and unique research topics under the broad umbrella of physical crystallography!

Ben Tragheim, Sheffield

PCG Plenary Lecture

The PCG Plenary lecture was chaired by **Lewis Owen** (Sheffield) and was given by **Abbie McLaughlin** (Aberdeen) who spoke on the subject of “Hexagonal perovskite derivatives: a new platform for the discovery of fast oxide ion conductors.” The speaker emphasised that in view of climate change there is much need for improved solid oxide fuel cells (SOFC) which can operate at lower temperatures than current devices such as 8 % yttrium-stabilised zirconium which operates at 800 °C. The speaker described her work on the design of solid ceramic electrolytes for SOFC’s including the use of fluorites and bivanadates which operate at lower temperatures as well as perovskites that work at around 600 °C.

Hexagonal perovskites exhibit multiple polymorphs with mixed conductivity that has an ionic contribution of less than 25 %. Titanates were studied as well as perovskites containing barium, niobium and molybdenum which were found to exhibit significant oxide ion conductivity. These effects were studied by impedance spectroscopy and neutron diffraction using the GEneral Materials (GEM) powder diffractometer at ISIS along with bond valence site energy calculations using SoftBV. The speaker’s research showed a disordered distribution of the two metal ions with intrinsic O vacancies as well as indicating how the structure of these materials varies with temperature. As temperature rises the number of tetrahedral centres (which are associated with greater conductivity) increases while the octahedral centres decrease in number. Hence it was decided to increase the vanadium content to increase the number of tetrahedra. High pressure neutron studies were conducted using Pearl at ISIS showing that pressure reduces the number of tetrahedra and that in Palmierite there is disorder in the barium and niobium positions and that increasing humidity increases conductivity, suggesting the involvement of protons. The speaker described the use of *ab initio* MD and neutron diffraction to find where the protons or deuterons and water molecules were going in the structure. This work suggested that the waters were entering a vacant oxide site and that conductivity is due to the oxide ions as well as protons, making it a mixed ion conductor.

Work on epitaxial thin films of barium and niobium which have high conductivity was described. Combining the results of neutron powder diffraction work at Polaris (ISIS) with Arrhenius plot studies of palmierites substituted with Sr and V showed that rotation of the vanadate group was very important for ion transport. Similar studies also showed that La can substitute for Sr and that lead doping gives rise to high oxide conductivity as well as high humidity-dependent proton conductivity with a clear deuterium isotope effect. The speaker attributed these effects to a softer lattice in which the lone pair pushes the lead ion off centre.

Abbie then described neutron studies of composite materials of barium, titanium and molybdenum using D2B at ILL, looking at interstitial sites and demonstrating that static disorder becomes dynamic at high temperature. The speaker ended by introducing a new class of perovskite material made from barium, tantalum and vanadium which has significant oxide conductivity.

Jon Cooper, UCL
Mark Montgomery, Syngenta



*The PCG Plenary Lecturer **Abbie McLaughlin** (Aberdeen) with chair **Lewis Owen** (Sheffield).*

Photographs by **Rebecca Clulow** (Uppsala), **Natalie Pridmore** (Durham) and **Tony Bell** (Sheffield Hallam).

CCG Plenary

The CCG Plenary lecture was given by **Neil Champness** (Birmingham) with the title “Complex-bearing metal-organic frameworks: snapshots of reactions.” Neil led us through a highly engaging and interesting journey of his group’s work investigating the structures and unusual properties of reactive components in MOFs. He began with transition metal-functionalised carboxylate linkers, in which exposure to light leads to isomerisation that can be seen with time-resolved spectroscopy and single crystal X-ray diffraction. The unusual aspect here was that the photochemistry in the solid state was opposite to that found in solution.

Single crystal methods could also be used to resolve the bound triazole reaction products and stages of the catalytic cycle in CLICK reactions between azide ligands and various alkynes within another MOF. The CO release and changes to the metal coordination environment in the same MOF could also be tracked in a similar way. Neil then told us about recent research into disassembling MOFs using metal-linker reactivity to form otherwise unobtainable frameworks with reduced dimensionality. The speaker also covered the modelling of correlated disorder in a MOF with a vacant binding site and multivariate lanthanide MOFs containing up to 16 different d-block metals. It was clear throughout Neil’s lecture that teamwork and collaboration played an important part in the research, which was a rich and varied display of the fascinating interplay between crystallography and chemical reactivity.

Hamish Yeung, Birmingham



Hamish Yeung (Birmingham) chaired the CCG Plenary lecture given by **Neil Champness** (Birmingham).

CCG Open Session

Yaser Balmohammadi (Cardiff) presented a talk entitled “A quantum crystallographic protocol for general use and its application in the YLID test crystal to reveal a hidden story.” In the lecture, which was chaired by **Fanny Costa** (Leeds), a practical framework that integrates quantum mechanical calculations with conventional single-crystal X-ray diffraction was described.

The motivation for the work lies in the limitations of standard independent atom model (IAM) refinements, which provide reliable atomic positions but often fail to describe electron density accurately. Full charge-density studies, while more informative, typically require high-resolution data that are not routinely available. The proposed quantum crystallographic protocol bridges this gap by combining theoretical wave functions with experimental diffraction data.

The workflow consists of resolution-limited data refinement, standard IAM refinement, visual inspection of residual electron density, followed by Hirshfeld atom refinement (HAR) and reassessment of residual density maps. Importantly, the protocol is simple, reproducible, and applicable to measurements collected at room temperature without the need for ultra-high resolution data.

The method was demonstrated using YLID (2-dimethylsulfuranylidene-1,3-indandione), a well-known benchmark crystal structure. Despite its extensive study history, the protocol revealed previously unreported features, including subtle electronic effects and evidence of chirality-related distortions.

The presentation concluded that quantum crystallography can be made accessible for routine crystallographic practice, enabling new insights even in well-established systems.

Lewis Jackson (Kent) presented a study on how crystallisation temperature influences spin-crossover (SCO) behaviour in co-crystalline materials. Speaking under the title of “The effect of crystallisation temperature on an abrupt spin-crossover co-crystal” the speaker described how spin crossover involves a reversible transition between low-spin and high-spin states in response to external stimuli. This is accompanied by changes in metal-ligand bond lengths, unit-cell volume, and colour which makes SCO compounds attractive for applications including sensors, molecular switches, data storage, and solid-state refrigerants.

The work focussed on SCO co-crystals, where crystal packing plays a central role in determining transition behaviour. Using the SCO-active complex and small monotopic hydrogen-bonding co-formers with varied steric and electronic properties, the study aims to establish structure – property relationships and enable rational materials design. Six crystal structures were solved by single-crystal X-ray diffraction, revealing competition between the solvent and the co-former inclusion.



Speakers in the CCG Open Session: Giulio Lampronti (Cambridge), Yaser Balmohammadi (Cardiff), Lewis Jackson (Kent) and Yevheniia Kholina (Oxford).

Three SCO responses were observed: abrupt, irreversible and gradual transitions. Notably, an extended tensor network enhances communication pathways through the lattice, enabling cooperative behaviour consistent with a wine rack-type mechanism. These results demonstrate how controlled crystallisation and design can tune SCO functionality.

Next, **Giulio Lampronti** (Cambridge) presented how time-resolved *in situ* powder X-ray diffraction tracks microstructural evolution during mechanochemical processes. The talk was entitled "Microstructural monitoring of mechanochemical processes via *in situ* XRD." The speaker outlined strategies for collecting high-quality *in situ* data during ball milling, including optimising beam geometry, refining instrumental resolution and mitigating issues such as peak overlap, absorption, and low signal-to-background ratios.

Building on these experimental strategies, Giulio demonstrated how diffraction data can be analysed beyond phase identification, extracting crystallite size, microstrain and defect accumulation to reveal mechanistic signatures of milling-induced transformations. Case studies included non-displacive polymorphic transitions, such as theophylline – benzamide systems under liquid-assisted grinding and amorphisation processes in MOFs (e.g. ZIF-4 and ZIF-62). These examples showed how induction periods, comminution limits and strain build-up govern phase transformations and loss of crystallinity.

This presentation showcased *in situ* PXRD as a powerful tool to connect macroscopic milling conditions with microscopic structural changes, offering deeper mechanistic insight into mechanochemistry and supporting the rational design of solid-state reactions.

Yevheniia Kholina (Oxford) then presented an insightful talk on "Responsive disorder in framework materials." She began by explaining how disorders can evolve from being

random to being correlated and how these correlations can be controlled to tune properties such as crystallinity and stability. Examples included framework corrugation induced by flash-freezing, irreversible vacancy reshuffling upon drying and large-scale framework reorganisation triggered by solvent exchange and gas loading.

A key methodological advance was 3D- Δ PDF analysis, which disentangled diffuse-scattering contributions in real space, enabled statistical refinement of correlated disorder and provided insights into switching pathways and intermediate states. Case studies on Prussian blue analogues and the flexible MOF DUT-8 showed how responsive disorder governs mass transport, framework flexibility and guest adaptation. Overall, the talk demonstrated responsive disorder as a powerful design principle, extending its relevance beyond gas storage and separation to emerging concepts such as rewritable information storage in framework materials.

Fanny Nascimento Costa, Leeds

CCG Session: Advanced Techniques for Molecular Crystals

This session was chaired by **Lauren Hatcher** (Cardiff) and was an engaging and wide-ranging session on "Advanced Techniques for Molecular Crystals." It highlighted recent developments across X-ray, electron and hybrid approaches to the structural analysis of dynamic and functional materials.

The keynote lecture was given by **Florian Kleemiss** (RWTH Aachen), who presented a lecture on "Fast quantum crystallography for spin-state determination by X-ray diffraction." Florian showcased the latest advances in non-spherical atom refinement, focusing on the NoSpherA2 platform within Olex2, which aims to bring quantum mechanically informed crystallographic refinement into routine practice. Through examples employing the HAR approach,

he demonstrated clear improvements in model quality even for datasets without ultra-high diffraction resolution and summarised systematic studies identifying computationally efficient yet accurate theory level and basis set combinations suitable for various small-molecule systems. A notable highlight was the implementation of effective core potentials to dramatically reduce wave function computation times. He concluded by illustrating how HAR enables confident assignment of electronic spin states in Fe(II) complexes, using aspherical residual density features to distinguish low- and high-spin configurations.

Our invited speaker, **Benjamin Palmer** (Bristol), spoke on “Highly refractive molecular crystals in animal coloration and vision.” His talk explored how many organisms exploit organic biocrystallisation, particularly of guanine, to engineer optical responses through precise control of crystal morphology and facet expression. He described how additives and dopants can guide crystal growth to produce highly refractive structures optimised for biological visual and signalling functions. Professor Palmer also introduced his group’s emerging work on biomineralisation in algae, a direction with potential ecological significance in understanding and managing phenomena such as algal bloom formation.

The third speaker, **Stuart Hayes** (EuroXFEL), presented on the subject of “Ultrafast diffraction with high-energy electrons: bridging ultrafast dynamics to atomic-resolution crystallography.” Dr Hayes provided a comprehensive overview of ultrafast electron diffraction (UED) as a tool for tracking atomic-scale motion across diverse phases of matter. He outlined the recent upgrade of the REGAE (Relativistic Electron Gun for Atomic Exploration) electron source at the European XFEL, which now enables both UED and 3DED modes to combine high temporal resolution and atomic-resolution structural completeness. Dr Hayes illustrated these capabilities with examples of ultrafast structural dynamics of I^{3-} and concluded by presenting some of the latest results obtained with REGAE.

The final talk was given by **Natalia Koniuch** (Leeds), who presented a lecture entitled “In-situ tracking of the microscopic crystallographic dehydration pathway in a model channel hydrate, theophylline.” Using low-dose scanning electron diffraction, Natalia has resolved the nanoscale structural evolution of this widely used pharmaceutical, whose commonly dispensed monohydrate form readily dehydrates under low humidity. Her work shows that water loss begins

at crystal surfaces where hydrogen-bonded channels terminate, with progressive dehydration exposing elongated monohydrate pillars in the parent lattice orientation that serve as sites for the nucleation of the anhydrous form. These results provide direct crystallographic validation of mechanisms previously inferred from bulk studies and offer insights that could inform improved stability, dissolution behaviour and overall drug bioavailability.

The session prompted interesting discussion and underscored the range of contemporary developments in crystallographic research. Many thanks to all speakers and attendees for their contributions to a productive and engaging session.

Lauren Hatcher, Cardiff

Microcrystals Analysis Workshop

The workshop session “Micro Crystals Analysis” aimed to provide an informative and educational overview of what can be achieved when working with very small crystals. The session showcased the current capabilities of different experimental methods, radiation types and instrument designs, highlighting their respective advantages, limitations and complementarities. Through a keynote lecture and a further two invited talks, followed by a panel discussion comprising all three speakers, this session sought to increase awareness within the community of the opportunities associated with microcrystal analysis, particularly addressing the challenges of samples that lie at the limits of conventional crystallographic techniques.

The session was chaired by **Daniel Rainer** (NCS, Southampton) and featured three speakers whose expertise spanned electron diffraction, synchrotron-based microfocus X-ray diffraction and emerging small molecule serial crystallography approaches. The keynote talk was delivered by **Arianna Lanza** (Copenhagen) who opened the session with her talk entitled “Electron crystallography: how a wild beam can be tamed to become a reliable multi-purpose tool.” She provided an overview of modern 3D electron diffraction (3D ED) methods and described how advances in instrumentation, data collection strategies, complementary analyses and processing workflows have transformed electron crystallography into a reliable technique for structure determination in a way analogous to single-crystal X-ray diffraction. Arianna emphasised the unique strengths of electron diffraction, particularly for materials that are inaccessible to conventional X-ray methods, while also stressing the considerations required for the choice of instrumentation, experiment design and data interpretation.

The second talk, “Bright beams and microcrystals: the future of structure determination at Diamond” was delivered by **Anna Warren** (DLS). Her presentation focussed on the capabilities of Diamond’s micro- and nano-focus beam lines for analysing small crystals, particularly in the context of macromolecular crystallography. Anna discussed how high-brilliance synchrotron radiation, optimised optics and specialised environments enable data collection from crystals that would previously have been considered too small for conventional XRD techniques, including standard synchrotron beam lines. Attention was also given to instrument design, with an emphasis on workflow integration and the practicalities of working routinely at cryogenic temperatures.

The final talk: “A picture is worth a thousand diffraction spots – using photometry to perform rapid, sample-versatile



Speakers in the CCG Session on Advanced Techniques for Molecular Crystals: Natalia Koniuch (Leeds), Lauren Hatcher (Cardiff, chair), Florian Kleemiss (RWTH Aachen), Benjamin Palmer (Bristol) and Stuart Hayes (EuroXFEL).

serial crystallography” was presented by **Ben Coulson** (DLS). This talk introduced the concepts, sample preparation and instrumentation currently being implemented at the I19 beam line at DLS to make serial crystallographic experiments available to a wider user base. Inspired by projects involving photo-crystallography, crystals can now be deposited directly on a plate substrate and using a photometric approach to determine candidate crystals, the plate is scanned and diffraction data collected sequentially. The image-based pre-selection of the crystal was highlighted as being able to streamline experiments by identifying suitable crystals rapidly and efficiently as well as calculating the fastest trajectory for data acquisition.

Following the talks, a panel Q&A session with all three speakers generated lively discussion and addressed several practical and conceptual issues. A recurring theme was that, despite considerable technical advances, many of these techniques remain “methods of last resort” for particularly challenging samples. As Arianna noted, there will still be materials that do not diffract, even under optimal conditions, because these methods are often used to probe the limits of measurability. Questions on experimental timescales highlighted that the duration of a 3D ED experiment depends strongly on how it is defined and on factors such as the instrument used, the scientific question, operator experience and crystal symmetry.

The panel also discussed the challenges of data selection and merging. While 3D ED experiments often involve measuring large numbers of crystals, identifying a single “best” dataset is not straightforward and depends on multiple quality metrics. In some cases, merging data may improve results, while in others it may not be beneficial. Practical advice was also offered, including the continued importance of investing time in crystallisation. Despite advances in micro-crystal techniques, improving crystal quality remains fundamental to successful structure determination across all methods.

Other questions included the contribution of the background, which was identified as a key factor affecting the signal-to-

noise ratio and transferring samples from one characterisation technique to another. Anna noted that such sample transfer is already being heavily considered in the workflows at Diamond, with instruments such as VMXm and HEXi designed with cross-instrument compatibility in mind. While not all techniques allow direct transfer of samples, the panel agreed that sharing strategies, methodologies and lessons learned across all approaches continues to benefit the whole field of diffraction as a whole.

Overall, the session successfully demonstrated how complementary micro-crystal techniques are expanding the boundaries of modern crystallography, while reinforcing that careful experimental design, fundamental crystallisation practices and crystallographic expertise remain as important as ever.

Daniel Rainer, NCS Southampton



*The CCG workshop session on “Micro crystals analysis” was chaired by **Daniel Rainer** (NCS Southampton) and included speakers **Ben Coulson** (DLS), **Arianna Lanza** (Copenhagen) and **Anna Warren** (DLS).*

Fellow Membership of the BCA

Applications are invited for fellow membership of the BCA via the members area of <https://www.crystallography.org.uk/>

Fellows of the BCA shall have an established career in crystallographic teaching or research and must hold one other class of membership.

Fellows will normally have been members for at least five years, but the BCA shall consider exceptions such as those with an established career abroad or members who have taken recent career breaks.

In addition to recognising an established career, fellow membership provides a simple way to support the association. The current rates for fellow membership are set at double those of the member's normal renewal price.

Science Day 2026: recognising student research, collaboration and scientific discovery

The CCDC supervisors and wider team organise the annual Science Day meeting to welcome [our sponsored students](#) and hear about the latest developments in their research. It is a day to listen, learn, and share knowledge, while also making time for socialising and friendly conversations.

This year, we heard from students in the final stages of their studies and welcomed a few new students. While it is always a little sad to see students move on, it is also incredibly rewarding to have witnessed their progress and to celebrate everything they have achieved. At the same time, our new students bring fresh ideas, exciting possibilities, and renewed enthusiasm for scientific discovery. We highlight the research and achievements that were presented during this year's meeting.

Alex Lee (Durham) is finishing his PhD which explores the use of crystallographic phenomena as alternative sources of chirality in organic synthesis. He investigates how conglomerate crystallisation can create enantiomerically enriched crystals from racemic materials, through processes such as Viedma ripening. By synthesising and screening 2-amino-4H-chromene derivatives, Alex identified factors that promote conglomerate formation and significantly improved discovery rates through a targeted screening approach. His work also shows that solid solutions can induce conglomerate behaviour in compounds that would not normally exhibit it, providing new opportunities for chiral enrichment and crystal engineering.

William Midgley (Durham) is researching computational PROTAC design. He has developed a workflow that combines docking, linker generation, conformer modelling and molecular dynamics simulations to identify PROTAC candidates capable of forming stable ternary complexes. A key focus of his work is optimising linker design and improving scoring methods to predict effective PROTAC conformations better. His research is currently being applied to EBNA1, a protein associated with Epstein–Barr virus-related cancers and multiple sclerosis, to accelerate the discovery and prioritisation of promising PROTAC therapeutics.

Aaron Horner (Southampton) is a fourth-year PhD student in Simon Coles' group. Currently in his last year, presenting at the Science Day annual meeting, Aaron mentioned what motivated him to start his PhD with CCDC: *"I had the opportunity to work as a summer student with CCDC, and found that I learned a lot during this period. It furthered my own interest in data science, and it felt like a good direction to take a PhD if I were able to. I'm fortunate to say I was."* Aaron's research focuses on assessing crystal structure quality, particularly for challenging crystal sponge structures where traditional metrics such as the *R*-factor are often

insufficient. He has developed automated tools, including the Distortion Score, Weighted Ratio and Bond Likelihood methods to provide more reliable measures of structural quality and identify problematic refinements. By enabling large-scale analysis of crystallographic datasets, his work supports improved data quality, standardisation and future applications in data-driven crystallography and materials discovery.

Sarah Madeira (Southampton) presented her work on crystal structure prediction (CSP) for organic photocatalysts used in sustainable hydrogen production. Her research investigates how crystal packing affects charge transport in organic photocatalysts. Using CSP, she successfully predicted experimental crystal structures for known photocatalysts and generated energy-structure-mobility maps to understand their performance. A key challenge is the high computational cost of calculating crystal structures and electronic properties.

Matthew Shaw (Leeds) presented his research on using machine learning to predict crystal habits. Crystal shape plays an important role in pharmaceutical manufacturing, affecting properties such as processability and solubility. Current prediction methods can be computationally expensive, so Matt is exploring data-driven alternatives using CSD data. His work involves building machine learning models to classify crystal habits as block, needle or plate using molecular and crystal structure descriptors. While initial models have shown limited accuracy, including 3D crystal structure information appears to improve performance.

Paulo Nunes (Sao Paulo) is focusing his research on improving the solubility and bioavailability of an anti-HIV drug through the development of new salt forms and co-crystals. Using computational screening, Paulo has evaluated a wide range of organic and inorganic co-formers, leading to the preparation of several new solid forms. He is currently characterising these materials using crystal structure modelling, charge density analysis and X-ray constrained wavefunction refinement to understand how structural changes affect key pharmaceutical properties. His goal is to establish structure–property relationships that support the design of more effective drug formulations.

Henry Holleb (Durham) is researching the prediction and control of hydrate formation in pharmaceutical salts. Using over 31,000 crystal structures from the Cambridge Structural Database (CSD), he investigates how the choice of counterion influences a salt's tendency to form hydrates. By combining crystallographic data analysis with electrostatic potential calculations, Henry has shown that charge localisation is a key factor governing hydration behaviour. His work identified clear trends across several families of counterions and enabled the development of predictive models that estimate hydrate propensity, even for systems with limited experimental data. He has successfully validated these predictions through experimental salt screening, demonstrating that counterions with more diffuse charge

distributions are less likely to form hydrates. His research provides a practical framework for selecting counterions and designing more efficient pharmaceutical solid-form screening strategies.

Henry James Broster (Oxford) is combining machine learning and physics to improve molecular docking and generate ligand-binding poses that are both accurate and physically realistic. His recent work uses reinforcement learning to teach diffusion-based docking models to optimise for physical validity, rather than relying solely on traditional measures such as RMSD. By incorporating physics-based criteria, including PoseBusters validation and interaction recovery, his approach produces higher-quality docking predictions and improves performance on challenging protein–ligand systems. This research demonstrates how machine learning and physics-based methods can be combined to generate more reliable molecular docking models, supporting drug discovery applications where both accuracy and physical plausibility are essential.

Omar El-Habbak (Strathclyde) is in the final year of his PhD working with CMAC in collaboration with AstraZeneca. His research focuses on using machine learning to predict pharmaceutical powder flowability from experimental and computational particle properties. By combining laboratory measurements of particle size, shape and flow behaviour with crystal structure descriptors generated from CCDC tools, Omar has developed predictive models that classify powders according to their flow performance. His work integrates data curation, crystal form matching, morphology prediction and synthetic data generation to improve model accuracy despite limited experimental datasets. As he enters the final stage of his PhD, Omar reflects on his journey as a CCDC-sponsored student: *“The most rewarding part was by far my CCDC placement, underscored by the warm, friendly environment and how quickly my project progressed in a short period of time from being around people who knew the CSD Python API inside and out.”*

This year, we welcomed a group of students from the University of Oxford who are part of the ILESLA programme,

a doctoral training programme led by the University of Oxford in partnership with Oxford Brookes University and The Open University. The programme provides interdisciplinary PhD training in the life and environmental sciences, combining taught coursework, rotation projects, and collaborative research opportunities across academia and industry. Two groups of students presented their short project research from the past year, carried out in collaboration with CCDC Discovery Science researchers.

Stanislavs Kurass is a first-year DPhil student with Elisa Villacampa Teixeira, Inka Snellman and Hannah Pitchford. As part of a team challenge project co-supervised by Andrei-Vlad Badelita and Kepa Burusco-Goni, they are exploring how machine learning can be used to create compact molecular descriptors bridging 3D structures and electrostatic potentials that better capture the properties relevant to protein–ligand binding and molecule similarity.

Naomi Costello is also a first-year DPhil student participating in developing a drug-like subset of the CSD by applying computational filters based on Lipinski’s Rule of Five and beyond. The project aims to identify molecules with properties associated with good oral bioavailability and to create a curated dataset to support drug discovery and molecular design workflows.

The Science Day meeting continues to be a valuable opportunity to connect with our students, recognise their achievements and explore the diverse ways crystallographic data and computational methods are advancing scientific research. From crystal engineering and pharmaceutical solid forms to machine learning and drug discovery, this year’s presentations showed the impact and relevance of the work being carried out by the CCDC-sponsored PhD students.

We want to thank all the students, supervisors and collaborators who contributed to this year’s event. We look forward to following the progress of our returning students, supporting those approaching the completion of their PhDs and welcoming the exciting discoveries that our newest researchers will bring in the years ahead.



Participants at the 2026 CCDC Science Day.

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Register your place now:

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6th October at 9:30–11:15 BST

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[Discover new Mercury features for crystal optimisation and interaction networks](#) **3rd November** at 13:30–15:15 GMT

Watch on demand

Using the CSD to teach chemistry: from high school to PhD

Some concepts in chemistry, such as symmetry elements in crystal structures, can be challenging for students to engage with. In this July webinar, we explored how educators use the Cambridge Structural Database (CSD) to support learning, helping students visualise and analyse molecules and crystal structures. The CSD is also widely used in outreach activities

to introduce crystallography to students who might not otherwise have the opportunity to explore crystal structures. Two educators from our community share how they use the CSD portfolio to teach chemistry across a wide range of levels, from high school students to PhD researchers.

Speakers:

Dr Erhard Imer, Lecturer for Chemistry at Georg August University of Göttingen and XLAB, Göttingen Experimental Laboratory for Young People, Germany

Professor Natalia Alvarez, Adjunct Professor of Inorganic Chemistry at the Universidad de la República, Uruguay

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Ana Macado, CCDC

The Peaceful Crystallographer – and other songs

Electric Voice Theatre & Conway Hall Ethical Society,
15th October 2026, 18:30–20:15, Conway Hall,
25 Red Lion Square, London, WC1R 4RL.



Join Electric Voice Theatre at Conway Hall for a singing celebration of Kathleen Lonsdale – the first female professor at UCL and one of the first female Royal Society Fellows. Defying all odds, her pioneering science led to breakthroughs in organic chemistry and medical treatments. Beyond science, Lonsdale was a prison reformer, women's rights activist and



pacifist, warning of the dangers of nuclear weapons and wealth inequality in her book *Is Peace Possible?*

The music, performed by Electric Voice Theatre, UC8 and the Clerkenwell Community Choir, reflects Lonsdale's eclectic life. Ranging through Irish folk song, music-hall, stirring anthems and ethical hymns the performance features works by Judith Weir and Cheryl Frances-Hoad, Lonsdale's contemporaries like Elizabeth Maconchy, Shena Fraser and Clara Novello Davies and Conway Hall's radical thinkers, Eliza Flower and Josephine Troup.

Following the performance, we will examine her history with historian Dr Patricia Fara and meet some of today's women in science at UCL in the year of its 200th anniversary. This event is part of Bloomsbury Festival 2026. Ticket price: £20/£14/£9.

<https://www.conwayhall.org.uk/whats-on/event/the-peaceful-crystallographer-and-other-songs/>



BLOOMSBURY
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20 YEAR ANNIVERSARY
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Bernal Lecture 2026

THE 2026 Bernal Lecture at Birkbeck in London was held on 13th May and was given by Prof Alison Noble CBE FRS FEng (Oxford) with the title of “Artificial intelligence in science: a perspective from medical ultrasound imaging.” This series of lectures is held in honour of Prof John Desmond Bernal FRS (1901-1971) who pioneered the use of X-ray crystallography in molecular biology and published extensively on the historical and social dimensions of science. He was a communist activist who in 1937 was awarded both an FRS and a chair of Physics at Birkbeck (Wikipedia, the free encyclopedia) before going on to found the Department of Crystallography there.

The Head of Biology at Birkbeck, Carolyn Moores opened the event before handing over to the Vice Chancellor Sally Wheeler who described Bernal both as a towering figure of 20th Century science and a committed opponent of fascism and nuclear weapons, as well as the President of the World Peace Council from 1958 to 1965. The reins were then handed back to Carolyn who introduced the speaker by outlining her many distinguished honours.

Alison began by describing how AI is transforming science with new capabilities and tools, having particular impact in structural biology, atmospheric science, materials discovery, computer vision and code generation. The need to reflect on the role of AI in science was recognised in a Royal Society report entitled “Science in the age of AI – how artificial intelligence is changing the nature and method of scientific research” which the speaker was commissioned to chair. Alison emphasised that since AI relies on databases of research results which we all know represent the hard work of many thousands of people, it is essential to ensure the safe and ethical use of AI in scientific research.

The speaker moved on to discuss the role of AI in transforming radiology, the impact it has on the workplace and skill sets as well as human trust, data sharing and access, in addition to reporting standards and the requirement for AI regulation. The speaker described how things are moving so fast that we need to timestamp everything and discussed a range of questions such as whether AI needs to be as good as a human expert and whether we understand the human skill set sufficiently well to be able to compare it with AI fairly. Should AI be treated as an automation tool or an assistant or as an instructor or peer?

The speaker then moved on to describe the drivers of change in the medical ultrasound imaging field from the 90's when a typical hospital device occupied the space of a desk to the present day when miniaturised hand sets are already in use and will potentially be reaching the home consumer market in the next few years. The speaker described how the regulatory environment in which this work is undertaken varies from country to country. For instance, traditions in some parts of the world have resulted

in legislation that prevents parents from finding out the gender of the unborn child, every hospital fetal scan having to be lodged with the government.

The speaker described how ultrasound imaging is a challenging discipline in terms of human interpretation and this has presented something of a barrier to its wider adoption as well as making it a hard subject to train people in. The speaker described a UK-India research partnership for pregnancy risk assessment. Freehand scanning is only useful in a highly skilled setting so training for this project has focussed on structured scan protocols involving defined sweeps. A database of such scans has been used to train AI for gestational age estimation from ultrasound alone. Although the results in a clinical setting bore a linear relationship with those determined independently using biometry, the speaker emphasised that further workflow optimisation is required and that trust studies are necessary to convince medical personnel of the value of clinical AI. Indeed the recognition of the importance of AI in automated fetal biometry has led to the formation of a spin-out company (Intelligent Ultrasound) which was recently acquired by GE Healthcare.

The speaker described how AI performs as well as an experienced sonologist, achieving 95 % agreement with experts which is comparable with expert-to-expert agreement. The speaker then described research addressing the issue of why ultrasound detection is difficult to master which focusses on studying sonographers and utilising sonography data science. This involves using AI to assess the human skill in a task by eye-tracking studies of experienced sonographers, recording where they looked and the time taken. Indeed the computer is now able to predict where an experienced sonographer will look and where they will move the probe and this is useful for trainees who can be provided with voice guidance with the newly developed “Sonomate” system.

The speaker moved on to describe emerging areas in AI such as federated learning using decentralised data and human-AI collaboration with the aims of minimising workload and maximising trust.

In summarising, the speaker reflected on large scale data capture and deep learning as well as the importance of interdisciplinarity, team dynamics of clinical professionals and collaboration with psychologists for reduction of ergonomic load in this evolving sphere. Future challenges include the legal barriers to health data, cost, career paths for workers in the field, reproducibility and openness as well as the differences in publication practises across the field. This concluded a very interesting lecture which was, as usual, followed by a drinks reception in the evening.

Jon Cooper, UCL



The 2026 Bernal lecturer **Alison Noble** (Oxford) presenting on “Artificial intelligence in science: a perspective from medical ultrasound imaging.”
Photograph: **Shabir Najmudin** (City St Georges).

Past and present Birkbeck crystallographers attending the 2026 Bernal Lecture: **Martin Vickers**, **Jeremy Cockcroft** (UCL), **Dave Houldershaw**, **Richard Westlake**, **Jeremy Baum**, **Shabir Najmudin** and **Crystallography News** editor, **Jon Cooper**.
Photograph: **Shabir Najmudin** (City St Georges).



Fellow members of the BCA

Dr David Aragao, Diamond Light Source
Dr Edward Bilbe, Johnson Matthey
Dr Richard Birkinshaw, Walter and Eliza Hall Institute
Dr Jeremy Karl Cockcroft, University College London
Professor Simon Coles, University of Southampton
Dr Susan Crennell, University of Bath
Dr Alex Gibbs, University of St Andrews
Dr Lauren Hatcher, Cardiff University
Dr David Hughes, University of Cambridge
Dr Benson Kariuki, University of Cardiff
Dr Tony Keene, University College Dublin

Dr Mark Montgomery, Syngenta
Dr Lewis Owen, University of Sheffield
Professor Simon Parsons, The University of Edinburgh
Dr Helen Playford, Rutherford Appleton Laboratory
Dr Timothy Prior, University of Liverpool
Dr Georgina Rosair, Heriot-Watt University
Professor Mark R. Sanderson, Imperial College, London
Dr Jeremiah Tidey, University of Warwick
Dr David Walker, University of Warwick
Professor John Wallis, Nottingham Trent University

LSBC Summer meeting

THE meeting was held on Tuesday, 2nd June from 15:00-17:00 in the Clark Kennedy Lecture Theatre, Queen Mary Innovation Centre. The attendees were welcomed by the organiser, **Ben Stieglitz** (QMUL) who introduced the first speaker **Callum Haste** (ICR) whose lecture title was “Visualisation of stepwise derepression of TFIIH in global-genome nucleotide excision repair.” The speaker began by outlining sources of damage to DNA including oxidation, alkylation and UV radiation as well as the unintended side-effects of anti-cancer drugs such as cisplatin which act at the DNA level. The lecture focussed on the repair of bulky lesions in the chromosome. This repair is effected by endonucleases cleaving out an oligonucleotide containing the damaged bases and the resulting gap in the DNA is filled in by nucleotide excision repair (NER). The XPC complex has a critical role in recognising the DNA lesion and thus triggering NER. Indeed the initials XP stem from the severe DNA repair syndrome xeroderma pigmentosum which is caused by hereditary mutations of proteins in this complex. In addition, the UV damaged DNA binding protein complex (UV-DDB) has an important role in both recognition and repair of UV-induced DNA photolesions. The multi-subunit transcription factor TFIIH is also important since it is known to unwind DNA at the site of a lesion. TFIIH consists of a core complex and a dissociable module known as “CAK” which is short for CDK-activating kinase [1].

The TFIIH core complex contains the DNA helicase XPD, which serves to unwind the DNA during NER, and the DNA translocase XPB, which aids in promoter-opening in transcription and cooperates with XPD in DNA bubble-opening during NER [2].

The main question being addressed by the speaker's research was to determine how TFIIH is de-repressed in NER. To this end, the speaker described the use of affinity grids in which the surface of a biotinylated lipid monolayer on top of a gold/carbon grid supports a two-dimensional streptavidin lattice onto which the protein of interest sticks due to it having a biotin affinity tag. This system has numerous advantages for structure analysis of the protein of interest but requires that the signal due to the streptavidin lattice be subtracted in data processing. Particle picking by neural network yielded two structures of complexes of TFIIH, one of which, XPC-TFIIH, was resolved at 3.3 Å resolution. The use of multi-body refinement of this complex revealed an unknown peptide bound to the p62 component and this is believed to represent an autoinhibited state in which the DNA channel is partially closed and the ATP binding site of XPD is blocked. Further impressive cryo-EM structural studies suggests that XPA partially de-represses the complex which opens the DNA-binding site and allows it to engage with single-stranded DNA at the lesion site during NER.

The next speaker was **Greg Towers** (Queen Mary) whose lecture title was “Structural biology reveals what's special

about pandemic HIV.” Greg began by discussing how some virus strains are infective and some are not and whether we can use the molecular differences between them as a basis for drugging pandemic-causing viruses. Greg explained how eukaryotic cells have molecules which detect DNA in the cytosol and they signal inflammatory and immune responses. Hence current thinking is that when HIV infects a cell, instead of the capsid dissociating in the cytoplasm and the RNA being reverse transcribed in the cytosol, the viral RNA is actually reverse transcribed within the capsid which is then able to pass through the nuclear pore complex so that the viral DNA is released within the nucleus, bypassing the innate defence mechanisms. Fluorescence microscopy and cryo-ET along with molecular simulations indeed suggest that despite the capsid being around 1000 Å wide it is able to pass through the nuclear pore [3].

The speaker described that whilst some strains of HIV are not particularly infective in normal conditions, suppression of the interferon receptor (part of the host cell's innate immunity) can result in the virus becoming hugely infectious. The speaker described the protein CPSF6 (cleavage and polyadenylation specificity factor subunit 6) which plays an important role in nuclear import of the HIV-1 capsid and integration of the viral genome into the host cell DNA. The site of interaction between the capsid protein and CPSF6 is targeted by the Pfizer drug PF74 which is known to inhibit HIV replication by blocking nuclear entry of the capsid.

Each capsid hexamer has a conserved size-selective pore allowing the entry of nucleotides for DNA synthesis in the HIV-1 core [4]. Development of the Chromaclade package [5] for analysis of biologically significant sequence variability and lineage-specific adaptation suggests that certain mutations cluster in pandemic strains of the virus and these are associated with more dynamic capsids that allow increased import of nucleotides into the viral core. The development of inhibitors which bind to the capsid, some of which trigger the innate immune response due to DNA leakage, is now a very active line of research. One example is the newly approved drug Lenacapavir (Gilead Sciences) which is given by long-acting 6 monthly injections and binds to the same site on the capsid protein as PF74.

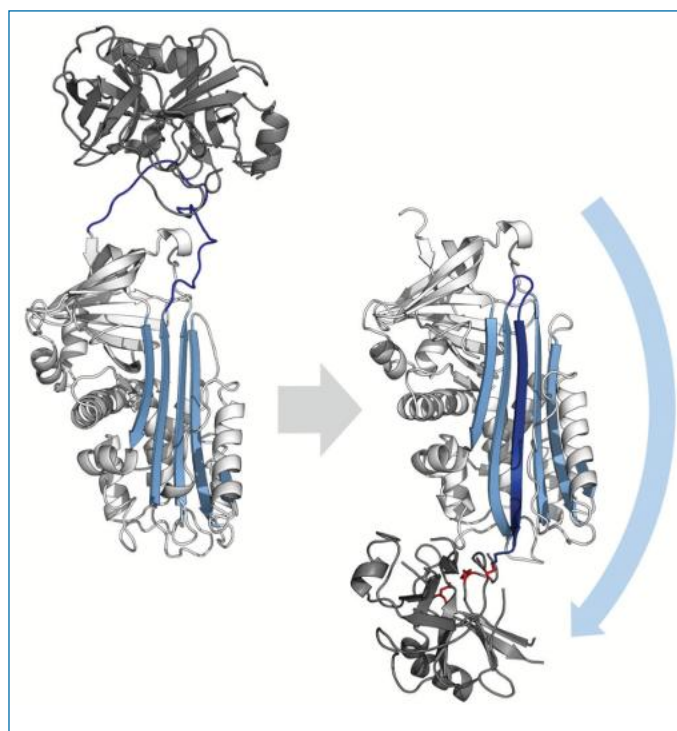
The next presentation was given by **Paul Driver** (Molecular Dimensions) who very kindly sponsored the meeting. Paul began by describing efforts to develop a vitrification screen for cryo-EM to try to minimise the problems of uneven ice, aggregation, preferred orientation and adsorption of the sample to the film (rather than it going into the holes in the grid) as well as adsorption to the air-water interface (AWI). The resulting screen consists of 19 conditions. Paul then described a new cocktail called surfACT which will include small LEA proteins (short for late embryogenesis abundant) from tardigrades which are known to withstand extreme desiccation stress. These LEA proteins, when used as additives during plunge freezing, have been shown to effectively mitigate AWI damage to fragile multi-subunit molecular samples [6].

The next invited lecture was given by **Mia Zol-Hanlon** (Francis Crick Institute) and was entitled “Finding the sweet spots: cryo-EM uncovers the role of glycans in RET neurotrophic signalling.” The speaker introduced the subject of growth control of neurons in the extracellular matrix, covering proliferation, differentiation, survival, guidance, migration and synaptogenesis as well as the involvement of glycans in these processes. The speaker moved on to glial cell derived neurotrophic factor (GDNF) which is a relatively small protein that potentially promotes the survival of many types of neurons and acts at the GFR α receptors, particularly GFR α 1 [1]. The latter is tethered to the outer membrane leaflet by a glycosylphosphatidylinositol (GPI) anchor. The complex of GDNF and GFR α 1 can interact with a tyrosine kinase called RET that is known to signal in both *cis* (affects same cell) and *trans* (affects other cells) and this forms the GFR complex. RET signalling is facilitated by glycosaminoglycans (GAGs), with heparan sulfate proteoglycans being particularly important [7]. Knocking out many components of the GFR1 affects kidney morphology in mice.

The speaker described the use of a neoglycolipid (NGL) microarray for screening for glycans which bind to GFR1 and the use of microscale thermofluoresis to validate the binders. Mia then described the structure of a complex of RET-GDNF-GTR α 1 with heparin determined by cryo-EM at 2.8 Å resolution which shows that the D1 cadherin-like domain of RET is engaged by the glycan. The glycan itself is long and spans over several of the proteins in the complex, analysis of which yielded better structural resolution for several of the protein components. The speaker also described an 8 Å resolution EM-derived model of a higher order complex of 4 GFR α 1 receptors which are held together by heparin. Mutation of heparin-binding residues observed in the model had no effects on *cis* activation but does affect *trans* activation, suggesting that GAGs facilitate the formation of longer range complexes than previously thought.

Last but not least, the final lecture was given by **James Irving** (UCL) and was entitled “Separating the beads from the string: the mechanism of oligomerisation in the serpinopathies.” The speaker began by reviewing the roles of serpins in neuronal development as well as blood clotting and the innate immune response. The serpin molecule possesses a reactive loop which binds to a serine protease and is cleaved by it. Whilst one side of the cleaved peptide bond is still attached to the serpin as an acyl-enzyme complex, the reactive loop of the serpin flips into a region of β -sheet converting it from a 5-stranded to a 6-stranded sheet. This conformational change pulls the catalytic serine away from the rest of the catalytic triad and the serine protease is inactivated by formation of the complex.

The speaker described how gram quantities of α_1 -antitrypsin are made by the liver per day to protect the lungs by inhibiting neutrophil elastase and this rate of production can increase 7-fold in an infection. Around 1 in 40 of the population are heterozygous for the Z allele (a mutation in the SERPINA gene) whereas around 1 in 2000 are homozygous and consequently suffer from emphysema, asthma, hepatitis and cirrhosis. The Z-mutant protein is prone to severe aggregation, forming fibrous deposits in which the serpin adopts the 6-stranded form even in the absence of the protease moiety. This form is thought to have inherently greater stability than the native 5-stranded form.



The reactive loop of the serpin is shown in dark blue with the serpin at the bottom-left and the target protease at the top-left in dark grey. Once cleaved by the protease, the reactive loop can swing down (as shown by the wide pale blue arrow) and insert into the serpin β -sheet which is formed initially by 5 pale blue strands. Insertion of the reactive loop gives a 6-stranded sheet and inactivates the protease by disrupting its active site. This image is from the Wikipedia [1] Serpin page and is available under the Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0/deed.en>).

Indeed the speaker spent some time describing NMR results confirming these findings. In the aggregates, the end of the reactive loop which becomes embedded in the 6-stranded sheet was shown to bind to another α_1 -antitrypsin monomer, thereby polymerising the monomers. Interestingly peptide mimics of the reactive loop can block this aggregation. The speaker described cryo-EM reconstruction of Fab conjugates of the fibrils which yielded a 4 Å resolution model of these beads-on-a-string aggregates.

The speaker concluded by describing similar studies of neuroserpin which can form inclusions in neurological tissues. Use of Fab complexes in a cryo-EM analysis revealed a 3.3 Å structure which was shown to be in the 6-stranded form. Studies are in progress for the development of polymerisation inhibitors (such as GSK71b) in collaboration with GSK under their rare diseases programme. The speaker mentioned that other mechanisms of aggregation are also possible.

The meeting closed with the regular networking session along with pizza and refreshments which were generously provided by Molecular Dimensions / Calibre Scientific.

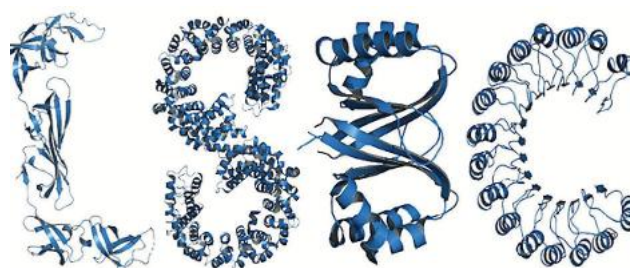
Jon Cooper, UCL



The rapidly-evolving East London skyline, as seen en-route to the London Structural Biology Club (LSBC) meeting which was held on 2nd June 2026 at the Queen Mary Innovation Centre.

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LONDON STRUCTURAL BIOLOGY CLUB

Pauline Harrison CBE (1926-2024)

After studying chemistry for one year in Edinburgh and then switching to Oxford (the year below Margaret Thatcher) Pauline Harrison (née Cowan) graduated in 1948 and pursued post-graduate research with Dorothy Hodgkin. In Dorothy's laboratory Pauline worked on vitamin B₁₂, virus protein crystals, peptides and gramicidins, as well as the iron-storage protein ferritin, receiving a DPhil in 1952. Subsequent post-doctoral work in the Biophysics group at Kings College London on fibre diffraction of collagen coincided with Rosalind Franklin's tenure there in the early 1950's. From 1956 to 1958 she worked in Sheffield on silicosis in miners and was then able to secure MRC funding to continue the X-ray analysis of ferritin as an independent researcher and later lecturer in the Department of Biochemistry. During this time she showed that the iron core of the protein was microcrystalline and by using EM and ultracentrifugation she showed that the iron-free form (apo-ferritin) can be reassembled from its constituent subunits.

Initially focussing on the structure and formation of the iron core, the project changed direction in mid-1970's when her group succeeded in determining the X-ray structure of apo-ferritin, demonstrating its detailed assembly from 24 subunits. The remarkably high symmetry of this protein means that it is now routinely used as a standard for cryo-EM single particle analysis (SPA) – indeed it has been resolved at near-atomic resolution using this technique.

Pauline held a Personal Chair of Biochemistry from 1978 to 1991 at Sheffield where numerous studies of ferritin and other proteins were continued by coworkers (including David Rice and the late Peter Artymiuk) amongst other collaborators. During her career she had numerous publications in Nature and was an invited speaker at many BCA meetings. She had major roles in the British Biophysical Society, acting as Chairman and Treasurer, and she also played a pivotal role in formation of the International Society for the Study of Iron in Biology and Medicine (BIOIRON).

Members are recommended to read the detailed obituary prepared by Simon Andrews (Reading) at the following URL: <https://bioiron.org/news/in-memoriam/in-memory-of-pauline-harrison.aspx>



Pauline Harrison CBE (1926-2024) held a Personal Chair of Biochemistry from 1978 to 1991 in Sheffield and conducted world-leading research on the structure of the iron storage protein ferritin. This photograph was very kindly provided by Sheila Harrison.

Sax Mason (1946-2026)

Sax Mason was born in rural Australia, in a small town approximately halfway between Melbourne and Adelaide and received a PhD from Queen's College at Melbourne University. He then began working in the neutron diffraction field at AINSE in Sydney before spending 3 years working with Dorothy Hodgkin and Terry Willis in Oxford and Harwell. Here he pursued some of the first neutron diffraction studies of insulin and lysozyme before moving to the Diffraction Group at ILL, Grenoble in 1973 where he continued work on these proteins. In Grenoble, Sax worked on a huge range of crystallographic projects which benefited from the high neutron flux at ILL and eventually the development of reliable 2D area detectors. These projects ranged from the structure of organometallic compounds to polysaccharides, proteins and DNA as well as enzyme mechanisms.

In conversations he always reminisced enthusiastically about the convivial atmosphere in Dorothy's laboratory in Oxford, conveying many unique and interesting observations of people in her group at the time. His wife Rosalyn Mason was head of the ESRF User Office until 2008.

For more details of Sax's many contributions to neutron science, readers are referred to the excellent recent obituary by Alan Hewat and Judith Howard in the IUCr Newsletter at the following URL: <https://www.iucr.org/news/newsletter/volume-34/number-3/sax-mason-1946-2026>

Miroslav Papiz (1955-2026)

Miroslav Papiz was a member of the UK's Synchrotron Radiation Source (SRS) team throughout its operational life from 1981 to 2008. He completed a PhD at Napier College, Edinburgh, working with Lindsay Sawyer on the structure of β -lactoglobulin, for which he achieved a first author paper in Nature. After this, he joined John Helliwell at Daresbury Laboratory as a postdoctoral research assistant for what proved to be the successful installation, commissioning and use of the Enraf-Nonius FAST TV electronic area detector on Station 9.6 of the SRS Wiggler Protein Crystallography Beamline.

In the following years, there were many outstanding publications from the user programme, including papers on virus crystallography from Purdue, Oxford and Harvard Universities. Miroslav also took part in regular user support at the Daresbury synchrotron, which was often an arduous task over many hours in the working day. The facilities presided over by Miroslav also contributed to another Nobel Prize-winning project, the structure solution of the ribosome, by another team from the MRC – LMB led by Venkatraman (Venki) Ramakrishnan.

After the closure of the SRS in September 2008, Miroslav moved to the University of Liverpool as an independent research scientist and continued his research work on light-harvesting complexes.

This text has been abstracted from the obituary of Miroslav which has been published by the IUCr [1] and can be found here: <https://journals.iucr.org/d/issues/2026/07/00/me6382/index.html>

I am grateful to two of the authors of the above obituary (John Helliwell and Pierre Rizkallah) for allowing me include the following figure which shows the extraordinary diversity of MX detectors installed and maintained by Miroslav on station 9.6 at the SRS over the years.

	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	00	01	02	03	04	05	06	07	08
7.2																											
9.6	Not built			Film			Film + FAST																				
9.5	Not built																										
14.1																											
14.2																											
10.1																											

Miroslav's PhD supervisor, Lindsay Sawyer (Edinburgh), points out that: "Miroslav was my first ever PhD student and it took me a while afterwards to realise that few were as good as he!"

Jon Cooper, UCL (Emeritus)

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Anthony Powell FRSC (1964-2026)

We regret to share the news that Professor Anthony Powell, a member of the Department of Chemistry at the University of Reading, died in May after a long illness. Anthony was an eminent solid-state chemist and a highly active researcher in thermoelectric materials, which convert heat into electrical energy. He led multidisciplinary teams exploring how these materials can be used to generate sustainable, low-carbon energy.

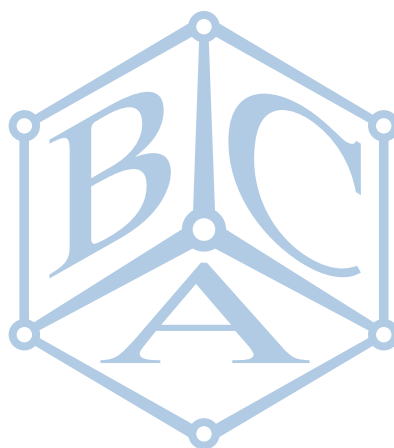
Anthony graduated with a degree in Chemistry from the University of Oxford in 1986 and went on to complete a PhD in 1990 under the supervision of Peter Dickens. His doctoral research focused on uranium-oxide insertion compounds.

Following his doctorate, he undertook postdoctoral research at Oxford, first with Tony Cheetham and then with Peter Battle. He joined Heriot-Watt University in Edinburgh as a Lecturer in Inorganic Chemistry, was promoted to Reader in 2000, and became Professor in 2008. Anthony joined Reading in 2013 as Head of the Department of Chemistry and Professor of Solid-State Materials Chemistry. He subsequently became Research Division Lead for Chemical Sciences and led our submission to the Research Excellence Framework (REF) for this Unit of Assessment.

Anthony's research interests encompassed thermoelectric materials, magnetic materials, microporous solids, and neutron scattering. He published more than 150 research papers, with more than half of those produced during his time at Reading involving international collaborations. Among his most recent projects was a Global Challenges Research Fund (GCRF) collaboration with the renowned Jawaharlal Nehru Centre for Advanced Scientific Research in India and the University of Manchester. The project sought to develop cost-effective technologies to convert waste heat from cooking stoves into electricity using materials that are readily available in India. Anthony's international standing in the field of thermoelectric materials led to his appointment to the Board of the International Thermoelectric Society and to invitations to deliver keynote and invited lectures at conferences worldwide. In addition, he served on an International Panel of Experts at Chongqing University, China, to advise on thermoelectric research, together with mentoring and delivering seminars to graduate students.

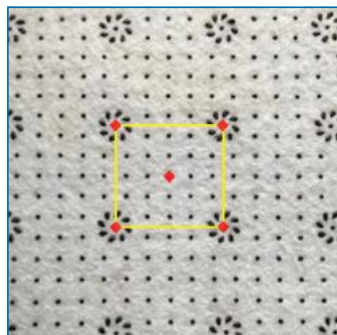
Anthony is remembered for his generosity with advice, especially to those early in their academic careers and his community spirit. His former research students have mentioned his patience, kindness and generous mentorship. We have received condolence messages from all over the world, including from former colleagues who described him as an outstanding chemist but also as thoughtful, kind and of admirable judgment and integrity. Anthony's family would be grateful for any donations to Cancer Research UK in his memory.

Paz Vaqueiro, Reading



Puzzle Corner

In the last issue we had four stunning patterns from **John Lisgarten** (London) for members to identify the plane groups thereof. I have had a stab at doing this myself, aided greatly by Volume I of the International Tables for Crystallography, my second hand copy of which dates from 1969 when it was published by IUCr in Chester.

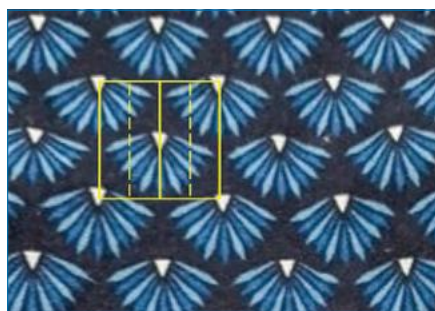


(a) After changing my mind about three times, I am convinced this is a gorgeous example of square $p4$ with the motif at the corners having local 8-fold rotational symmetry but, of course, only the four-fold symmetry extends across the lattice.



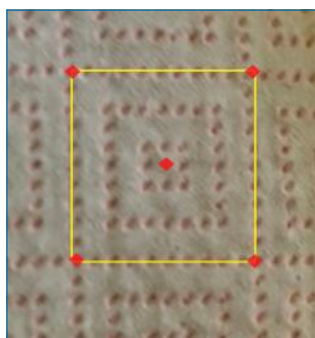
(b) The next one is a good example of $p3$ which the International Tables calls hexagonal, rather than trigonal, although I swear that I can only see three-fold axes here. Nevertheless I am not going to question the expertise that went into those hallowed tomes.

The three-folds are shown as red triangles and the mirror lines along the cells edges are shown as thick yellow lines, with the diagonal mirror line as a slightly thinner one.



(c) This peacock tail-like pattern is a nice example of rectangular cm with mirror lines along the vertical cell edges and through the middle. These mirror lines have

glide lines half way between them which are shown dashed.



(d) This one is an example of square $p4m$ with four-folds (drawn in red) at the corners and in the middle. There are numerous other diagonal mirror and glide lines which could also be drawn but these are probably best left for the interested reader to peruse in the International Tables.

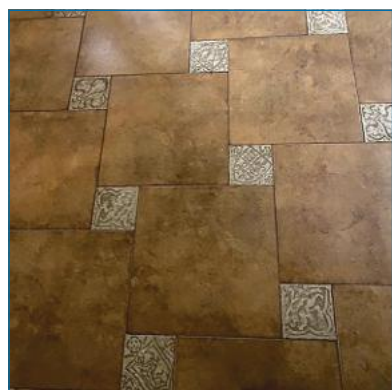
For this issue we have four more wonderful patterns from **John Lisgarten** (London) for members to consider in terms of plane group identification.



(a)



(b)



(c)



(d)

Meetings of Interest

WHERE possible, information on the following meetings has been abstracted from the conference websites, where further details may be obtained. Assistance from the IUCr website is also gratefully acknowledged. If you have news of any meetings to add to future lists, please send them to the Editor, jon.cooper@ucl.ac.uk.

Crystal36-AXAA 2026 Conference, Adelaide Hills, South Australia, Sunday 29th November to Wednesday 2nd December 2026

This is the 36th edition of the Crystals meetings of the Society of Crystallographers in Australia and New Zealand and the first time this meeting has been combined with a conference of the Australian X-ray Analytical Association. This will bring together the two organisations and potentially their combined 700 members for the first time, and will be a great opportunity to share knowledge and celebrate the achievements of these two groups.

We look forward to Crystal36-AXAA 2026 bringing together our two communities for a collaborative forum of discussion and debate on new and emerging issues, practical challenges encountered, experimental research findings, emerging trends and recent technological innovations and technical developments. The meeting will continue the strong traditions of both societies but create an environment for collaboration and cooperation.

Join us at the Adelaide Hills Convention Centre for the first joint meeting of SCANZ and the Australian X-ray Analytical Association. Hahndorf in the Adelaide Hills is a beautiful place to visit with lots to do and plentiful accommodation options onsite or a short walk away in Hahndorf. The venue is readily accessible from Adelaide Airport by public transport or other transport options.

More information is available at the meeting website: <https://www.crystal36axaa2026.org/>

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International SISN Neutron Science Conference (SISNSC 27), Firenze, Italy, 18th-21st January 2027

SISNSC 27 continues the longstanding tradition of the Italian Neutron Science Society (SISN) meetings in a fully international format. The conference aims to broaden scientific exchange and strengthen collaborations between the Italian neutron community and researchers from around the world who use neutrons as a research tool. The scientific program will be highly interdisciplinary, covering a wide spectrum of fields, from fundamental neutron research to the latest technological advancements.

We look forward to welcoming you to Firenze. Registration and abstract submission are now open, and we warmly invite you to register and submit your contributions through the conference website: <https://www.sisn.it/sisnsc/>

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Joint Polish-German Crystallographic Meeting, Dresden, Germany, 15th-18th February 2027

Dresden is one of Germany's and Europe's leading centers of research and innovation. With more than 40 research

institutions, TUD Dresden University of Technology, as one of Germany's eleven Universities of Excellence, and over 10,000 scientists, the city provides an outstanding environment for academic exchange, interdisciplinary collaboration, and technological progress. A particular strength lies in innovative materials research, spanning functional materials, nanoscience, quantum materials, microelectronics, and advanced characterization methods. Dresden also hosts a growing biomedical and life-science landscape, linking molecular, structural, and materials-based approaches to health, biotechnology, and medical innovation.

The city's microelectronics sector is especially strong: Dresden is widely regarded as Europe's largest and most dynamic semiconductor hub, with roughly one in three microchips manufactured in Europe coming from the Dresden region. With its exceptional density of scientific institutions, high-tech companies, and innovation-driven networks, Dresden attracts students, researchers, and specialists from around the world.

More information is available at the meeting website: <https://dgk-conference.de/>

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ECM36 2027, Prague, Czech Republic, 23rd-28th August, 2027

It is our pleasure to welcome you to the 36th edition of the European Crystallographic Meeting – ECM-36 which will take place in Prague.

ECM is a well-established conference series of the European Crystallographic Association (ECA, www.ecanews.org). Modern crystallography is an interdisciplinary subject covering structural biology, pharmacy, and chemistry to physics, and materials science. It can also be called structural science. It includes the study of protein structures and also crystals structures and microstructures of minerals, new functional materials, nanoparticles, bulk materials, and thin films. Important related topics are development of instrumental techniques for structural characterisation of materials including large facilities XFELs, synchrotrons and neutron sources, but also continuous improvement of laboratory equipment, development of optics, better and faster detectors. The core methods of crystallography are X-ray and neutron scattering. However, mainly in the last decade, it includes electron diffraction and microscopy, NMR and different spectroscopies. This development also allows investigation of structural changes under extreme conditions such as high and low temperatures, high pressures, time-resolved studies, different in-situ experiments in external fields (magnetic, electric) and chemical reactions.

The flagship of ECA activities is the European Crystallographic Meeting (ECM), held every year, except when there is an IUCr Congress. The ECMs are the main meeting point for crystallographers in the ECA area to show their recent research.

More information is available at the meeting website: <http://www.xray.cz/ecm36/>

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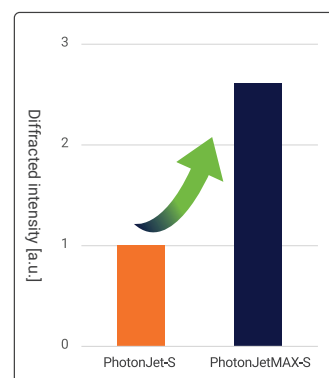
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